Learning a Codebase

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1 Glossary - Summary

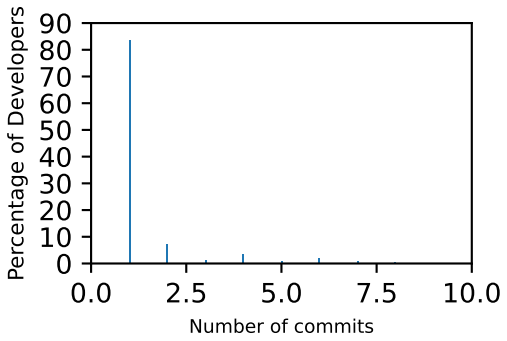
- A founder developer starts in the first six months of a project.
- A Later joiner developers start after six months.
- Transient developers have less than nine (9) commits.
- Moderate developers have more than nine (9) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

2 Repository Summary Table

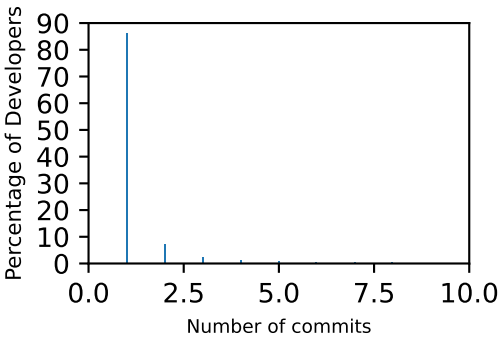
Table 1: Summary of thirty-six open-source repositories identified from GitHub Sustained later joiner developers who joined the project after six months and contributed at least 50 commits over a period of 250 days or more. Five developers each worked on two repositories.

ID	Repo Name	Transient Founder	Moderate Founder	Sustained Founder	Transient Later Joiner	Moderate Later Joiner	Sustained Later Joiner	Commit	Start	End
1	acs-aem-commons	3	0	0	189	0	0	1191	17-07-2013	Ongoing
2	acs-packaging	1	0	0	52	0	0	115	10-10-2014	23-11-2023
3	activiti	6	7	6	210	22	5	9836	18-06-2010	Ongoing
4	aem-core-wcm-components	1	0	0	100	0	0	366	06-01-2016	Ongoing
5	airbyte-platform	16	2	1	454	35	6	7002	04-08-2020	27-09-2023
6	alluxio	2	0	0	11	0	0	318	10-02-2013	01-12-2013
7	ambari	0	3	0	127	34	15	4357	22-09-2011	Ongoing
8	antlr4	3	0	0	293	0	0	3847	05-09-2011	Ongoing
10	automq	4	0	0	300	11	3	1889	10-08-2011	21-04-2017
11	avro	1	0	0	333	0	0	3339	15-04-2009	Ongoing
12	azure-sdk-for-java	8	0	0	11	0	0	1147	13-10-2011	08-01-2014
13	beam	57	0	0	38	0	0	6561	11-07-2014	20-05-2016
14	buck	17	6	6	524	92	21	15889	21-03-2013	05-05-2022
15	camel	1	4	1	1133	76	13	37463	19-03-2007	Ongoing
17	cas	2	0	0	305	0	0	46401	01-11-2010	Ongoing
18	checkstyle	0	0	1	359	45	9	17497	22-06-2001	Ongoing
19	commons-lang	4	0	0	202	0	0	3389	19-07-2002	Ongoing
20	cxf	6	6	3	201	17	3	8781	29-04-2008	Ongoing
21	intellij-community	0	3	7	334	103	84	104964	11-11-2004	18-04-2018
22	assertj	5	0	0	39	0	0	431	07-09-2010	09-04-2014
23	calcite	2	0	0	13	0	0	6211	20-04-2012	24-02-2014
24	dolphinscheduler	15	0	0	464	0	0	13029	29-03-2019	Ongoing
25	dubbo	4	0	0	176	0	0	4180	20-10-2011	07-04-2019
26	guava	0	1	0	400	10	3	9708	15-09-2009	Ongoing
27	hibernate-orm	4	0	0	15	0	0	531	29-06-2007	07-07-2010
28	jenkins	5	1	1	831	45	4	31546	05-11-2006	Ongoing
29	keycloak	3	0	0	1186	0	0	22954	02-07-2013	Ongoing
30	kubernetes-client	7	0	0	290	0	0	23224	09-07-2015	Ongoing
31	minecraftforge	2	0	0	493	0	0	20436	14-07-2011	Ongoing
32	openapi-generator	2	0	0	2646	0	0	30837	19-07-2011	Ongoing
33	openmrs-core	0	1	0	385	29	3	15769	03-05-2006	Ongoing
35	plugin-hub	18	0	0	347	0	0	8395	20-12-2019	Ongoing
36	presto	0	1	3	685	78	18	31858	09-08-2012	Ongoing
37	quarkus	2	3	5	820	29	8	82652	14-06-2018	Ongoing
38	runelite	0	0	0	673	0	0	60188	01-12-2014	Ongoing
39	selenium	5	3	0	709	20	8	23522	03-11-2004	Ongoing
	Total	206	41	34	15348	646	203	659823		

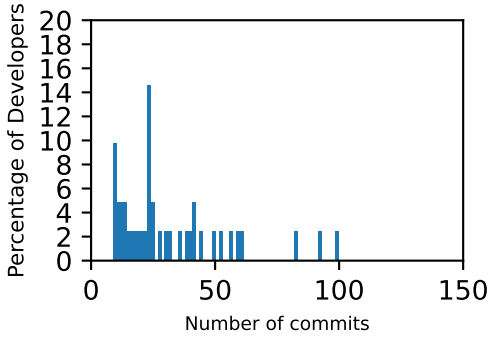
3 Total commits by developer - Developer Commits



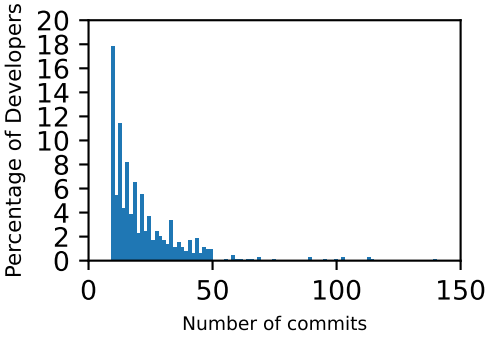
(a) Transient founder (n=206, y-axis max=83.5, x-axis max=8).



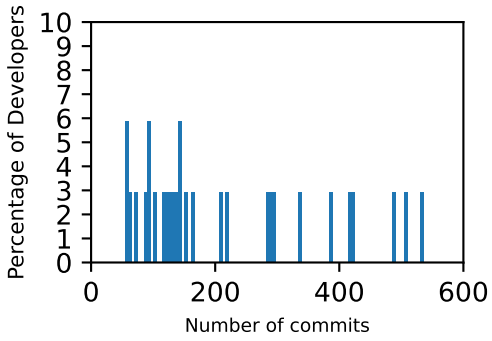
(b) Transient later joiner (n=15304, y-axis max=86.1, x-axis max=8).



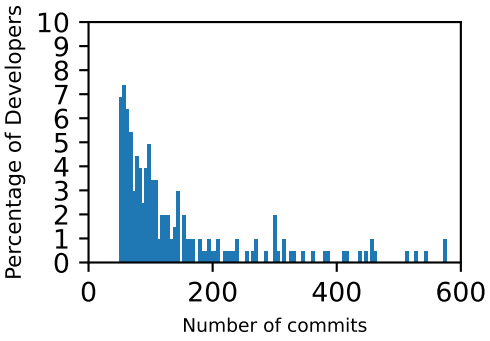
(c) Moderate founder (n=41, y-axis max=14.6, x-axis max=100).



(d) Moderate later joiner (n=646, y-axis max=17.8, x-axis max=304).



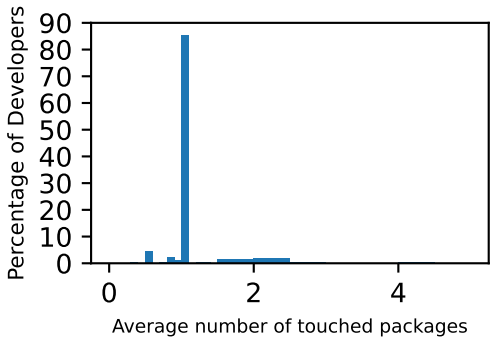
(e) Sustained founder (n=34, y-axis max=5.9, x-axis max=1255).



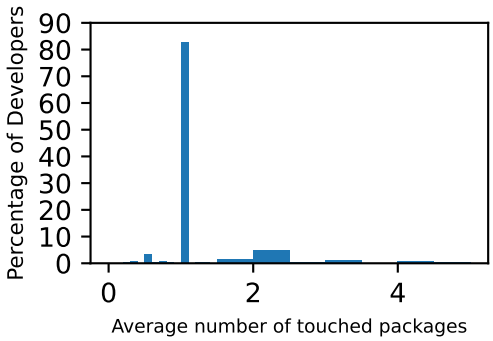
(f) Sustained later joiner (n=203, y-axis max=7.4, x-axis max=1062).

Figure 1: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 36 repositories sampled from GitHub.

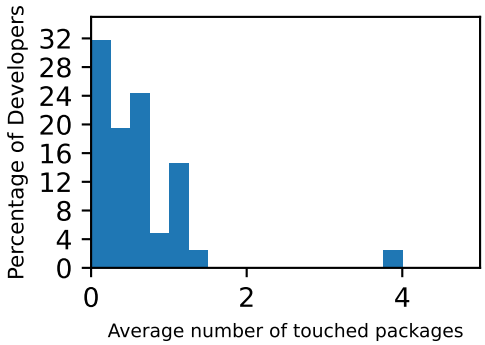
4 Histogram components touched by developer - For components with total



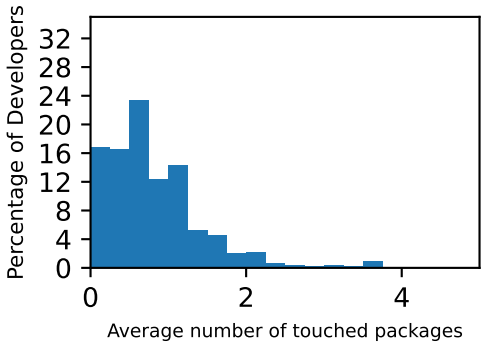
(a) Transient founder (n=206, μ=1.02, med=1.0, mod=1.0, max y=84.5, max x=6.0)



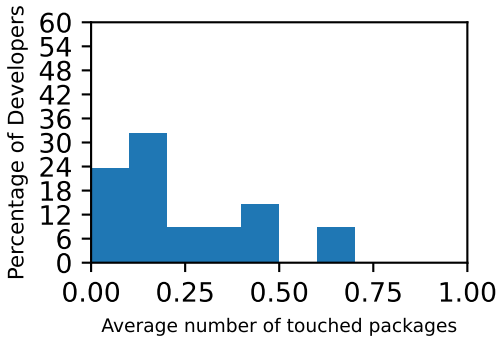
(b) Transient later joiner (n=15304, μ=1.07, med=1.0, mod=1.0, max y=81.6, max x=59.0)



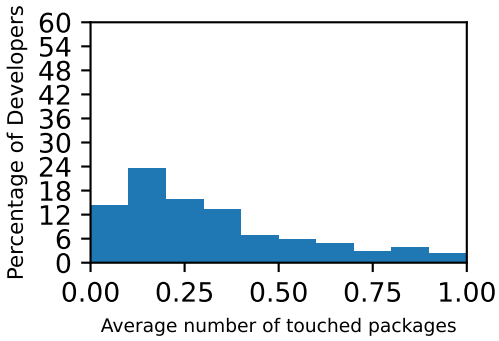
(c) Moderate founder (n=41, μ=0.13, med=0.1, mod=0.01, max y=39.0, max x=0.6)



(d) Moderate later joiner (n=646, μ=0.12, med=0.1, mod=0.01, max y=44.9, max x=11.0)

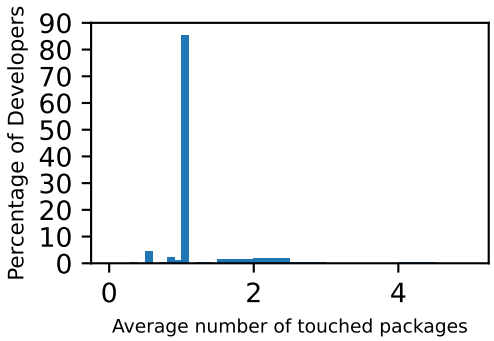


(e) Sustained founder (n=34, μ=0.1, med=0.1, mod=0.1, max y=44.1, max x=0.4)

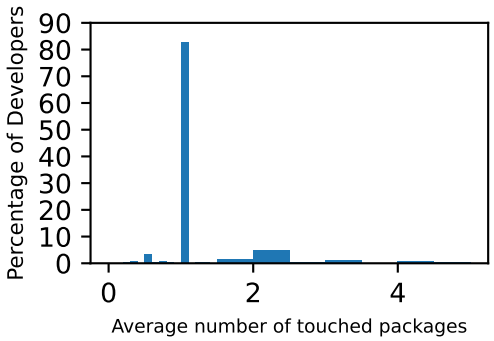


(f) Sustained later joiner (n=203, μ=0.11, med=0.1, mod=0.1, max y=41.4, max x=1.0)

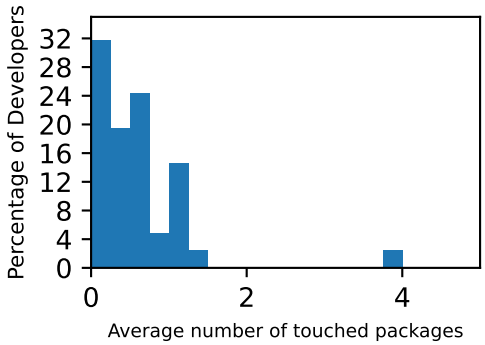
Figure 2: Histogram of percentage of developers (y-axis) against the average packages touched by daily (x-axis) from six (6) categories of developers from 36 repositories sampled from GitHub.



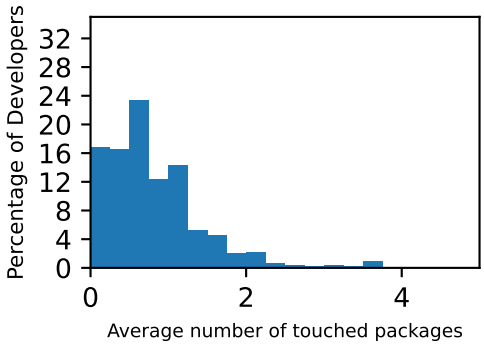
(a) Transient founder (n=206, μ=1.04, med=1.0, mod=1.0, max y=85.4, max x=6.0)



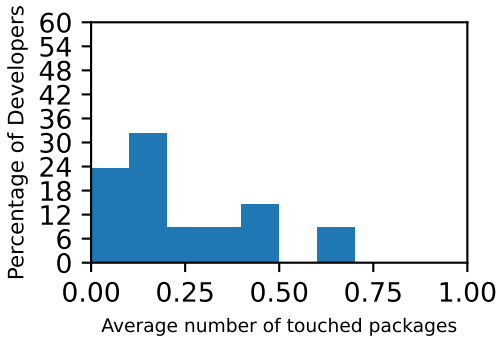
(b) Transient later joiner (n=15304, μ=1.15, med=1.0, mod=1.0, max y=82.8, max x=59.0)



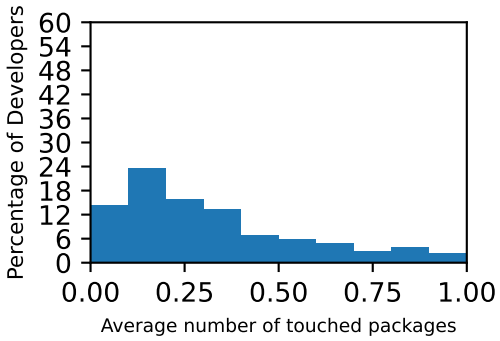
(c) Moderate founder (n=41, μ=0.58, med=0.4, mod=0.1, max y=31.7, max x=3.9)



(d) Moderate later joiner (n=646, μ=0.79, med=0.7, mod=0.4, max y=23.4, max x=5.4)

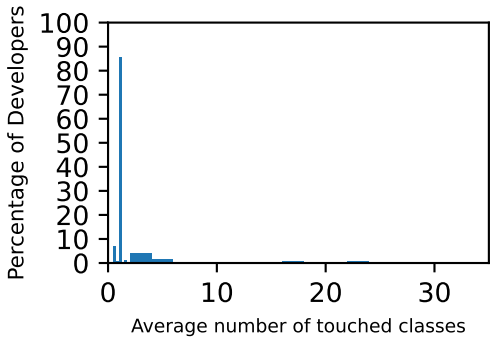


(e) Sustained founder (n=34, μ=0.23, med=0.1, mod=0.1, max y=32.4, max x=1.3)

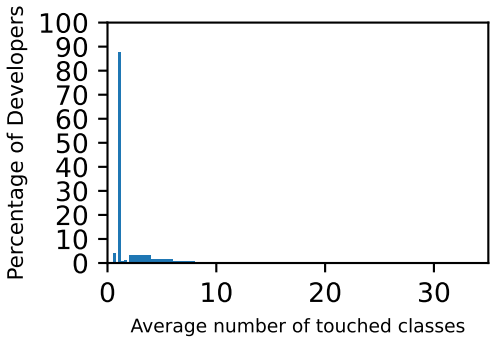


(f) Sustained later joiner (n=203, μ=0.36, med=0.2, mod=0.1, max y=23.6, max x=4.0)

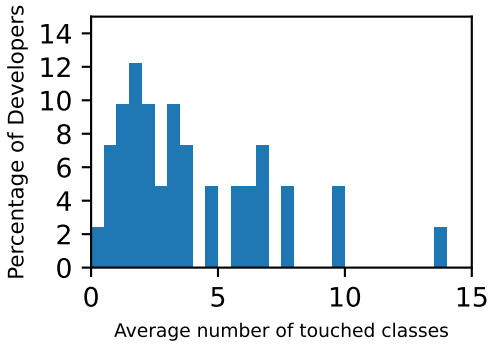
Figure 3: Histogram of percentage of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 36 repositories sampled from GitHub.



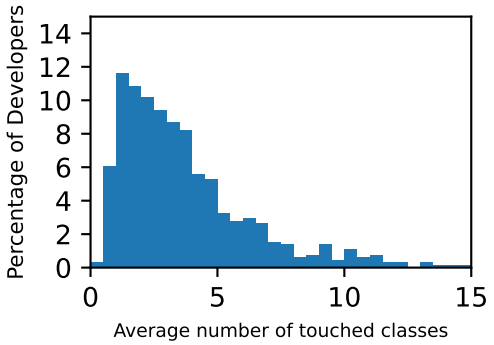
(a) Transient founder (n=206, μ=1.16, med=1.0, mod=1.0, max y=88.8, max x=35.0)



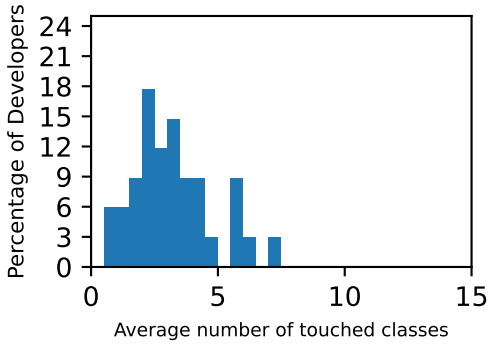
(b) Transient later joiner (n=15304, μ=1.05, med=1.0, mod=1.0, max y=88.3, max x=97.0)



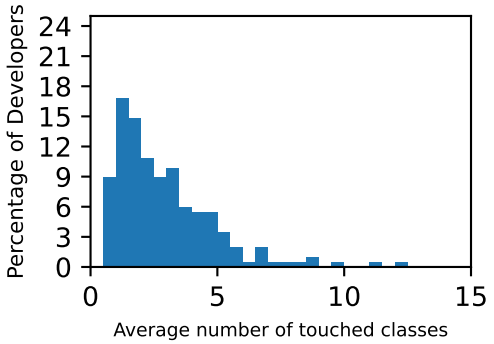
(c) Moderate founder (n=41, μ=1.1, med=0.7, mod=0.1, max y=39.0, max x=6.3)



(d) Moderate later joiner (n=646, μ=0.58, med=0.3, mod=0.1, max y=66.7, max x=15.0)

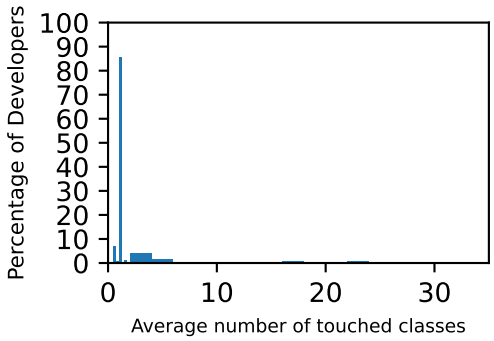


(e) Sustained founder (n=34, μ=2.31, med=1.9, mod=1.9, max y=17.6, max x=8.6)

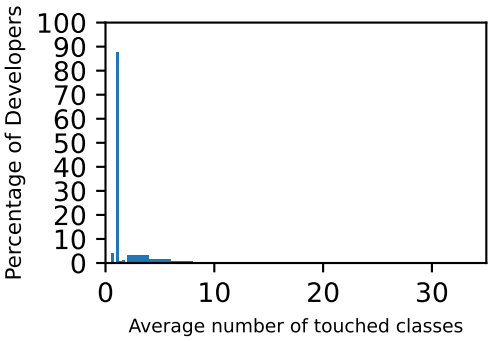


(f) Sustained later joiner (n=203, μ=1.32, med=0.9, mod=0.5, max y=36.5, max x=14.1)

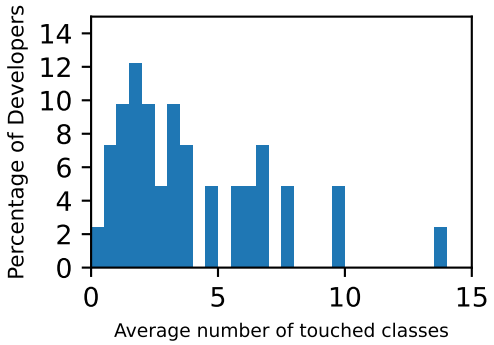
Figure 4: Histogram of percentage of developers (y-axis) against the average classes touched by daily (x-axis) from six (6) categories of developers from 36 repositories sampled from GitHub.



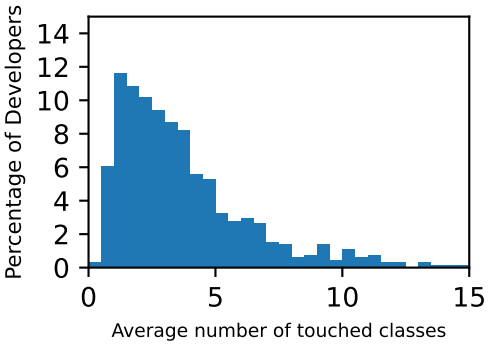
(a) Transient founder (n=206, μ=1.29, med=1.0, mod=1.0, max y=85.4, max x=23.5)



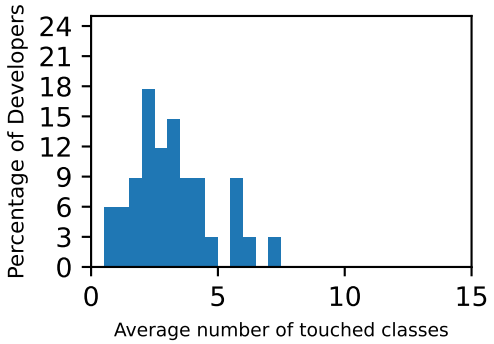
(b) Transient later joiner (n=15304, μ=1.3, med=1.0, mod=1.0, max y=87.8, max x=97.0)



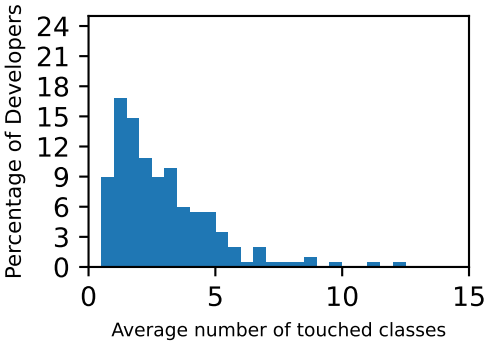
(c) Moderate founder (n=41, μ=4.69, med=3.0, mod=2.4, max y=12.2, max x=36.0)



(d) Moderate later joiner (n=646, μ=3.99, med=3.0, mod=1.1, max y=11.6, max x=38.6)

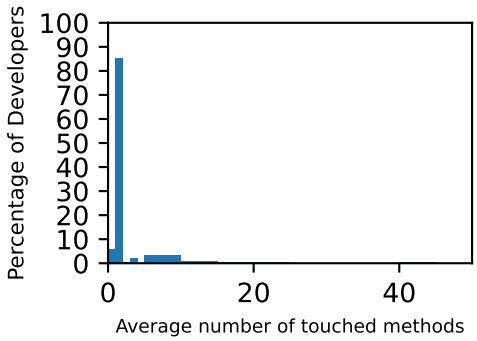


(e) Sustained founder (n=34, μ=3.14, med=2.95, mod=5.7, max y=17.6, max x=7.2)

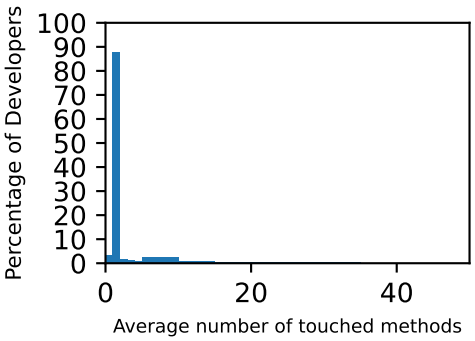


(f) Sustained later joiner (n=203, μ=3.19, med=2.4, mod=1.1, max y=16.7, max x=36.1)

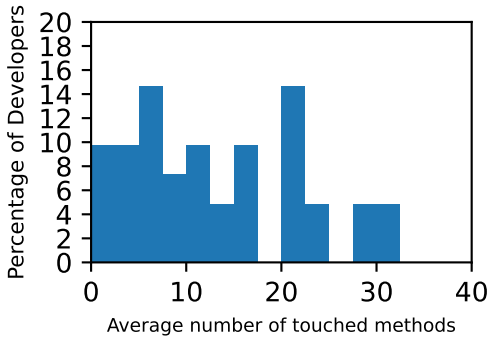
Figure 5: Histogram of percentage of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 36 repositories sampled from GitHub.



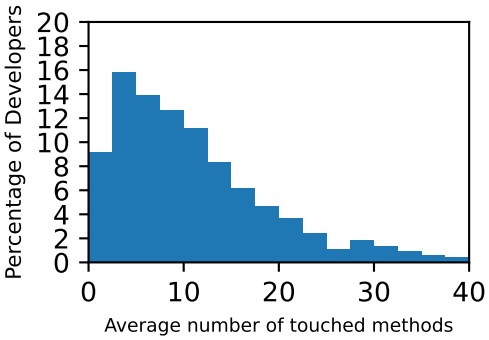
(a) Transient founder (n=206, μ=1.7, med=1.0, mod=1.0, max y=88.8, max x=109.0)



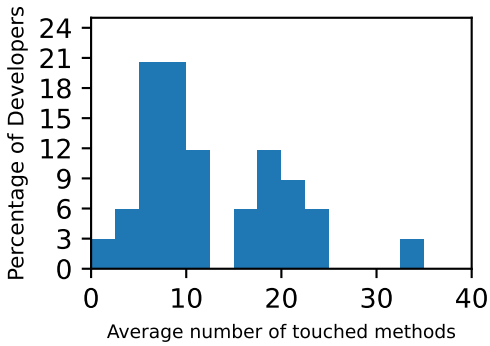
(b) Transient later joiner (n=15304, μ=1.37, med=1.0, mod=1.0, max y=88.9, max x=254.0)



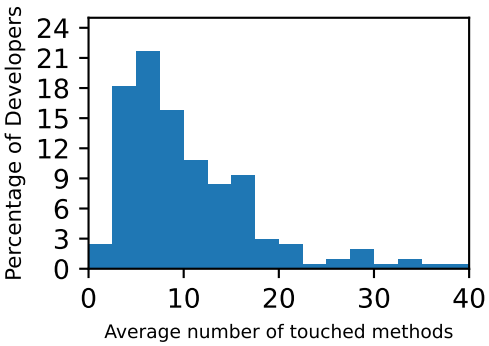
(c) Moderate founder (n=41, μ=4.45, med=2.4, mod=0.5, max y=51.2, max x=26.0)



(d) Moderate later joiner (n=646, μ=2.11, med=0.9, mod=0.3, max y=79.6, max x=76.3)

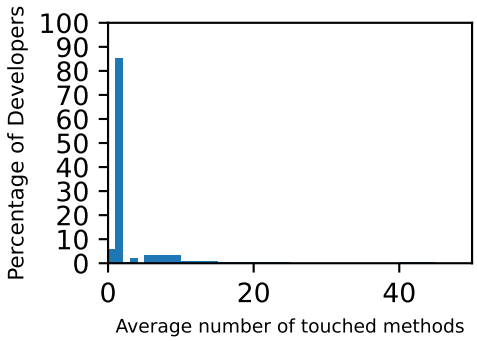


(e) Sustained founder (n=34, μ=9.61, med=7.7, mod=7.8, max y=20.6, max x=44.0)

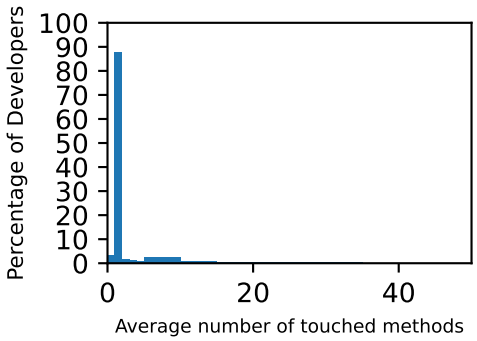


(f) Sustained later joiner (n=203, μ=5.02, med=3.1, mod=1.0, max y=41.4, max x=53.0)

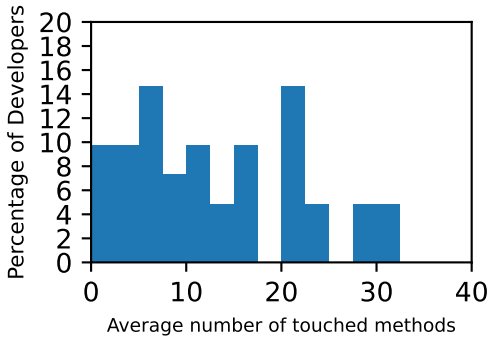
Figure 6: Histogram of percentage of developers (y-axis) against the average methods touched by daily (x-axis) from six (6) categories of developers from 36 repositories sampled from GitHub.



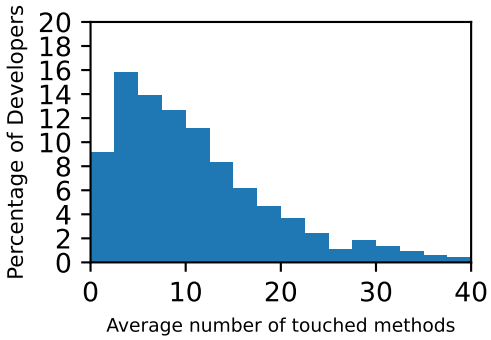
(a) Transient founder (n=206, μ=2.1, med=1.0, mod=1.0, max y=85.4, max x=76.0)



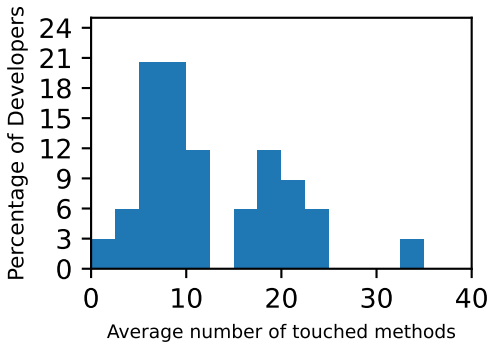
(b) Transient later joiner (n=15304, μ=2.23, med=1.0, mod=1.0, max y=87.7, max x=324.0)



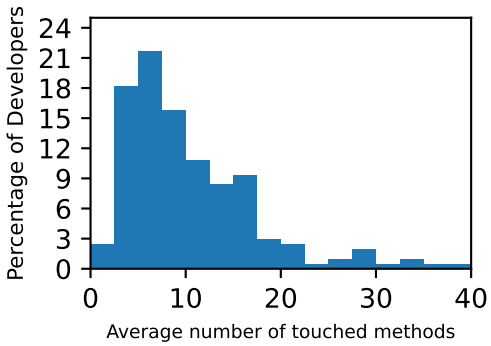
(c) Moderate founder (n=41, μ=20.6, med=12.4, mod=6.9, max y=14.6, max x=244.7)



(d) Moderate later joiner (n=646, μ=14.1, med=9.55, mod=6.2, max y=15.8, max x=155.9)



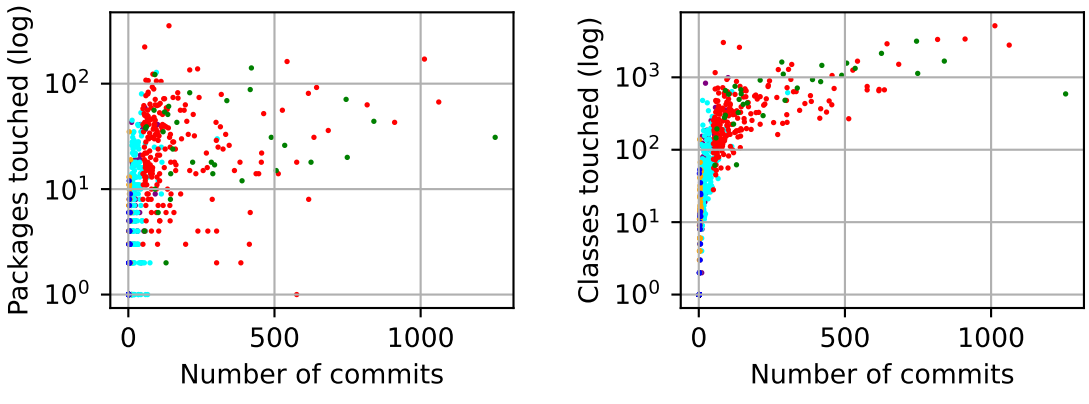
(e) Sustained founder (n=34, μ=13.28, med=10.15, mod=5.9, max y=20.6, max x=47.0)



(f) Sustained later joiner (n=203, μ=12.67, med=8.8, mod=5.5, max y=21.7, max x=182.2)

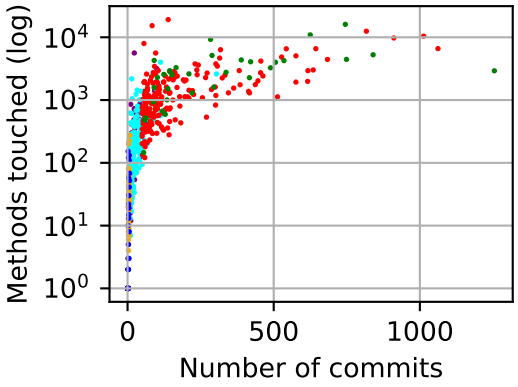
Figure 7: Histogram of percentage of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 36 repositories sampled from GitHub.

5 Sample scatter touched against commit -



(a) Packages

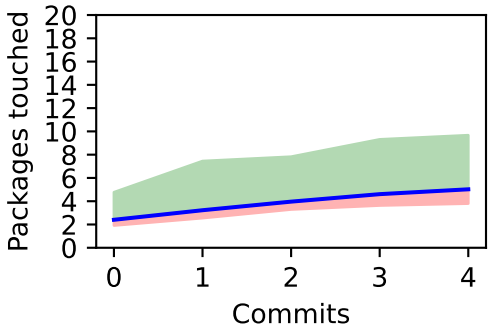
(b) Classes



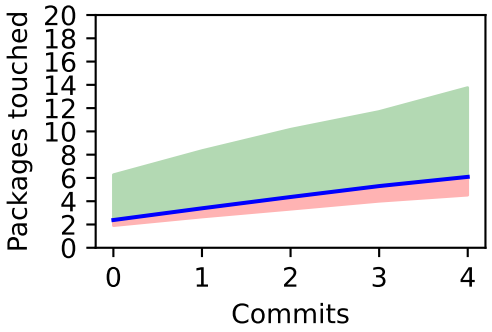
(c) Methods

Figure 8: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at 203 sustained later joiners from six (6) categories: Transient Founder (Blue, 203), Moderate Founder (Purple, 41), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

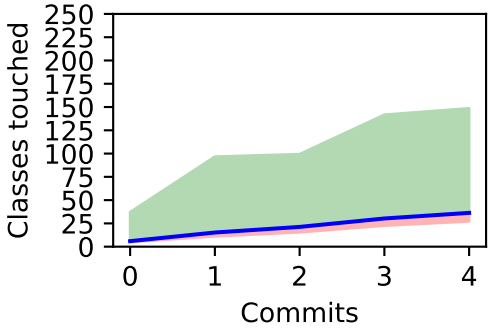
5.1 Time series components touched - For packages touched for first five commits



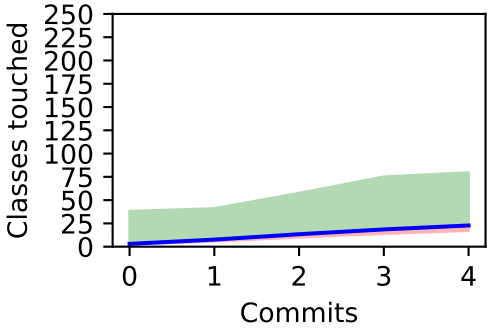
(a) Packages for All founders (n=84, μ=5.02, median=3.0 & std=3.57)



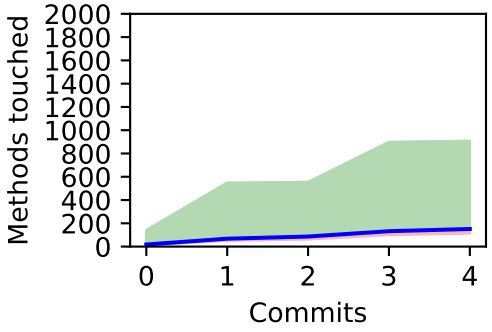
(b) Packages for All joiners (n=1277, μ=6.09, median=3.0 & std=4.66)



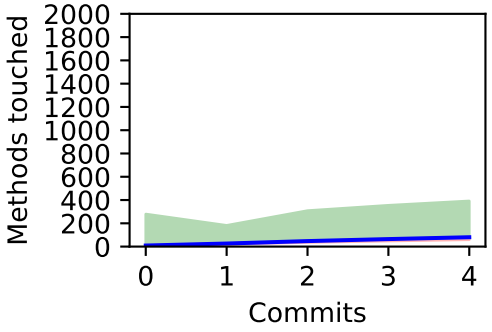
(c) Classes for All founders (n=84, μ=36.31, median=7.0 & std=51.51)



(d) Classes for All joiners (n=1277, μ=22.77, median=6.0 & std=28.25)

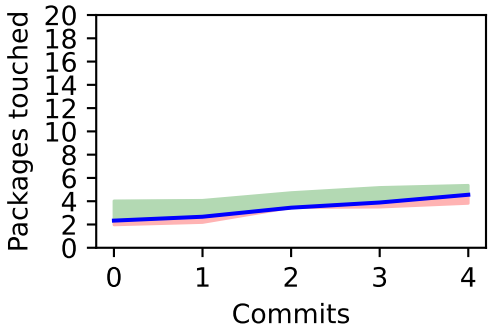


(e) Methods for All founders (n=84, μ=151.17, median=18.5 & std=294.09)

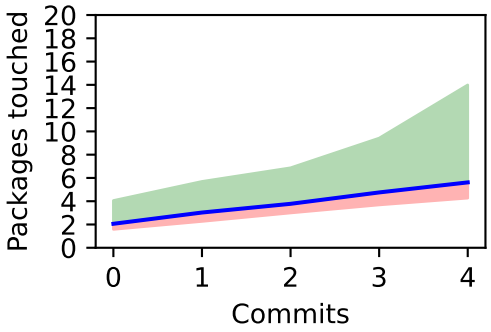


(f) Methods for All joiners (n=1277, μ=80.35, median=12.0 & std=138.17)

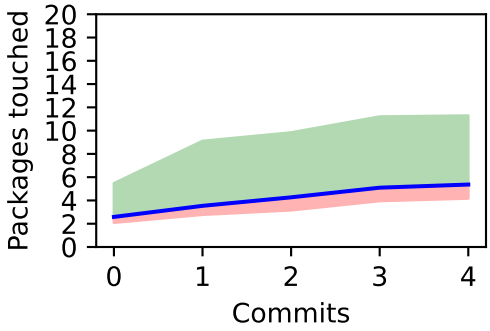
Figure 9: A time series for the five (5) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



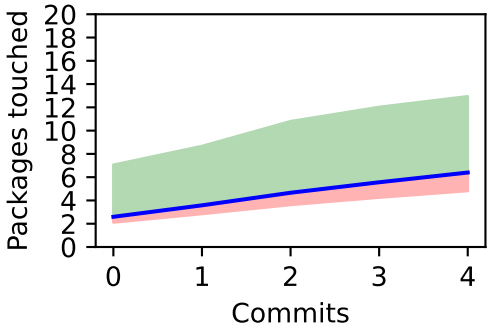
(a) Transient founder (n=9, μ=4.56, median=3.0 & std=1.99)



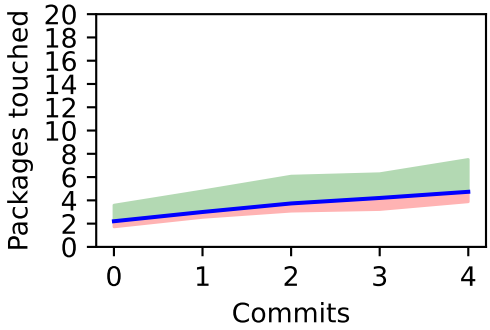
(b) Transient later joiner (n=429, μ=5.62, median=3.0 & std=3.96)



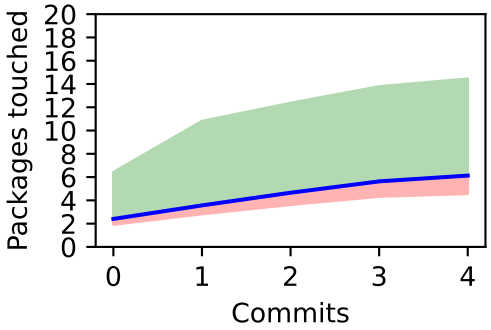
(c) Moderate founder (n=41, μ=5.37, median=3.0 & std=4.31)



(d) Moderate later joiner (n=646, μ=6.4, median=3.0 & std=4.82)

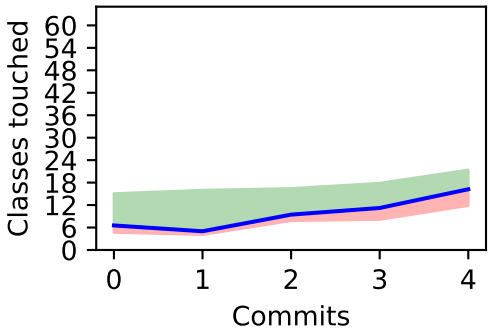


(e) Sustained founder (n=34, μ=4.74, median=2.0 & std=2.79)

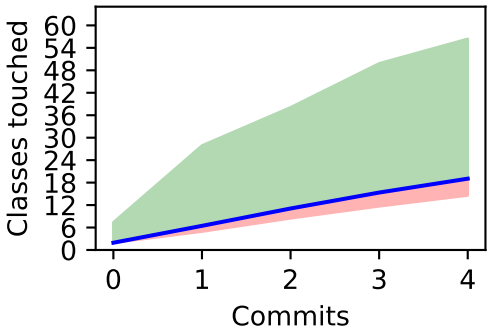


(f) Sustained later joiner (n=203, μ=6.13, median=3.0 & std=5.36)

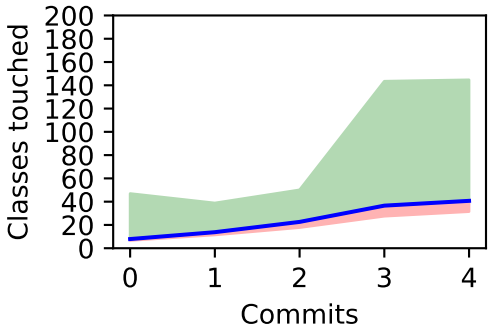
Figure 10: A time series for the five (5) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



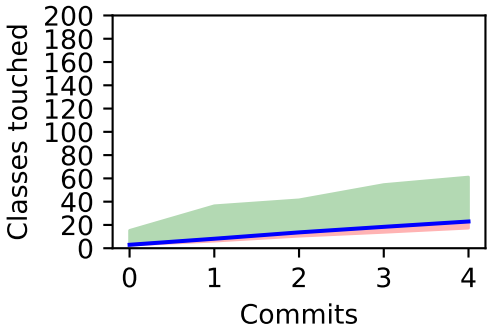
(a) Transient founder (n=9, μ=16.22, median=6.0 & std=9.9)



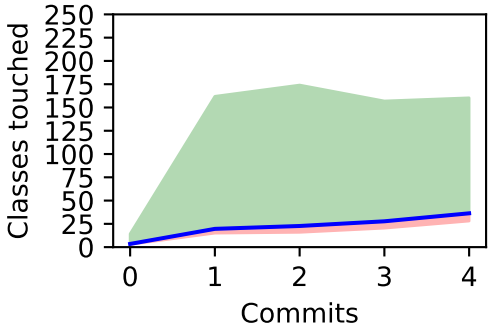
(b) Transient later joiner (n=429, μ=19.05, median=5.0 & std=19.33)



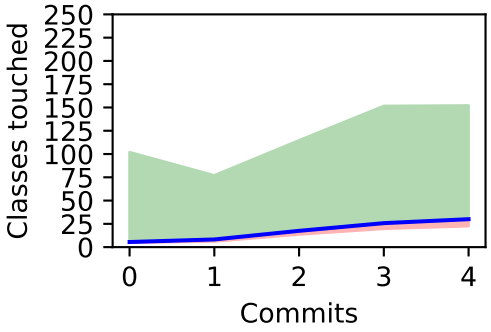
(c) Moderate founder (n=41, μ=40.71, median=9.0 & std=48.68)



(d) Moderate later joiner (n=646, μ=22.96, median=6.0 & std=21.93)

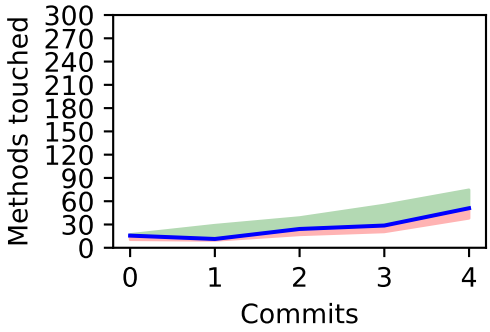


(e) Sustained founder (n=34, μ=36.32, median=7.0 & std=60.21)

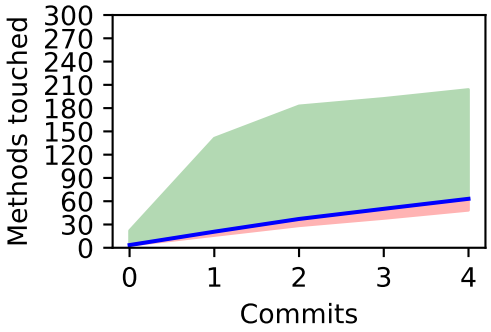


(f) Sustained later joiner (n=203, μ=30.01, median=6.0 & std=51.67)

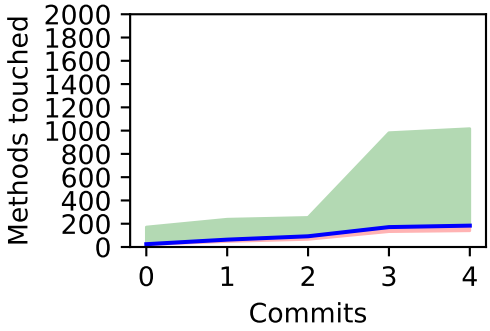
Figure 11: A time series for the five (5) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



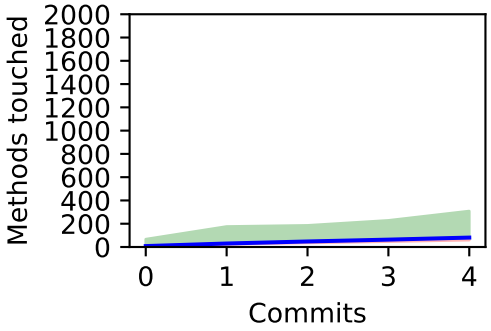
(a) Transient founder (n=9, μ=51.0, median=16.0 & std=30.34)



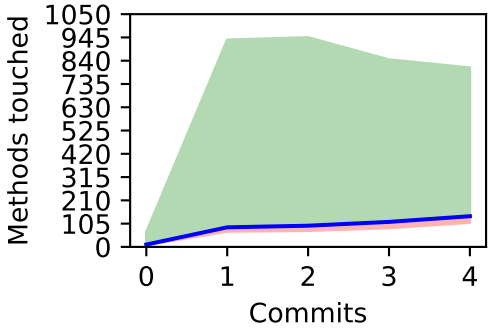
(b) Transient later joiner (n=429, μ=63.11, median=10.0 & std=79.44)



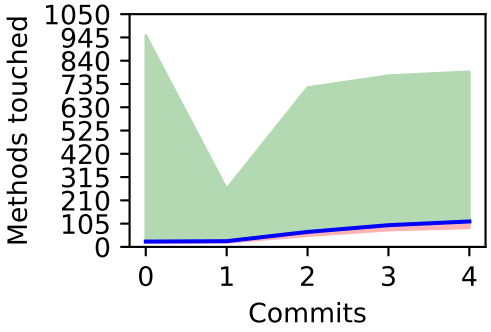
(c) Moderate founder (n=41, μ=183.63, median=23.0 & std=308.64)



(d) Moderate later joiner (n=646, μ=80.93, median=14.5 & std=100.84)



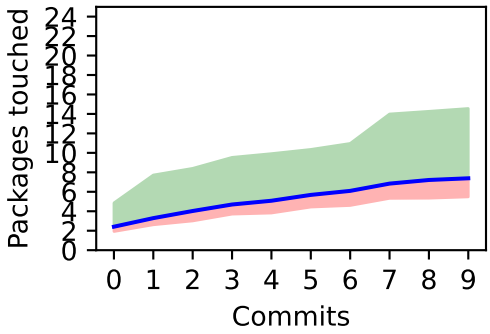
(e) Sustained founder (n=34, μ=138.53, median=15.0 & std=311.7)



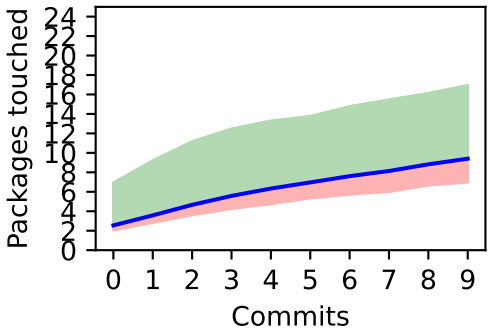
(f) Sustained later joiner (n=203, μ=114.77, median=13.0 & std=271.55)

Figure 12: A time series for the five (5) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

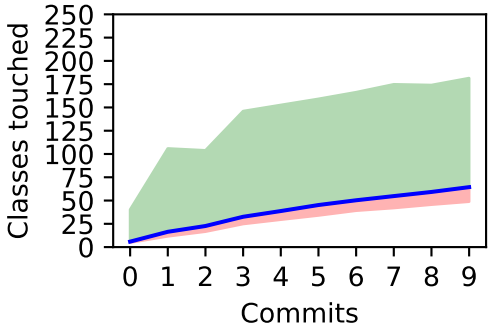
5.2 Time series components touched - For packages touched for first ten commits



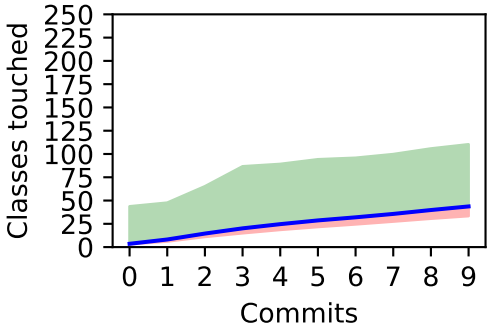
(a) Packages for All founders (n=75, μ=7.39, median=4.0 & std=5.2)



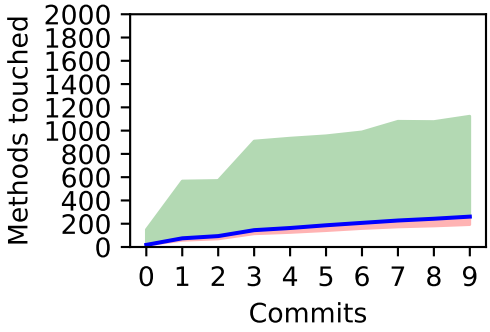
(b) Packages for All joiners (n=849, μ=9.4, median=5.0 & std=6.27)



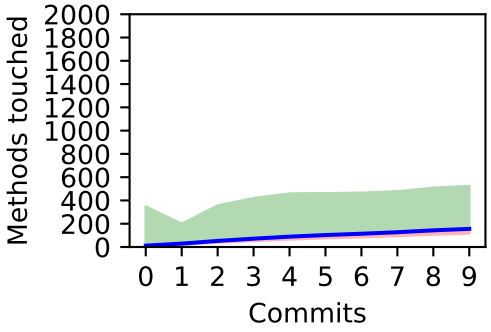
(c) Classes for All founders (n=75, μ=64.4, median=17.0 & std=70.95)



(d) Classes for All joiners (n=849, μ=43.67, median=14.0 & std=40.97)

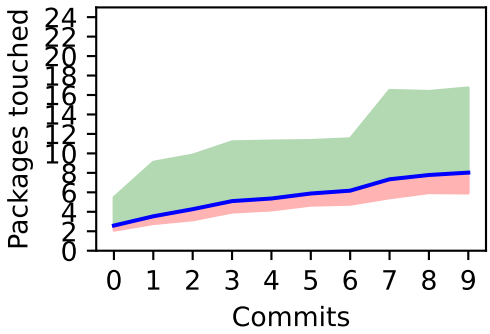


(e) Methods for All founders (n=75, μ=261.01, median=46.0 & std=414.75)

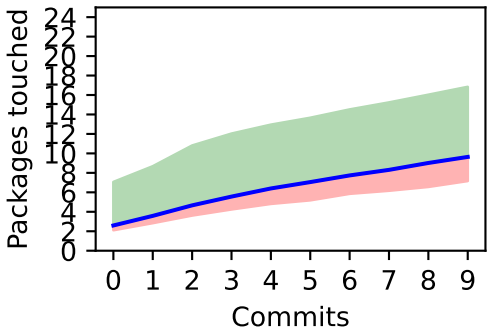


(f) Methods for All joiners (n=849, μ=155.65, median=37.0 & std=199.82)

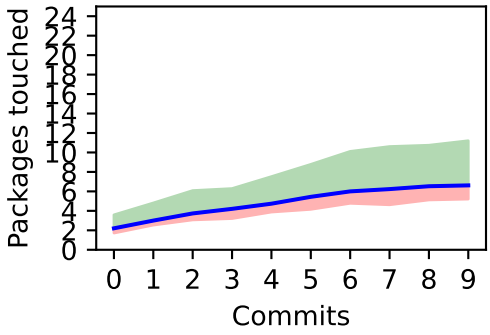
Figure 13: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



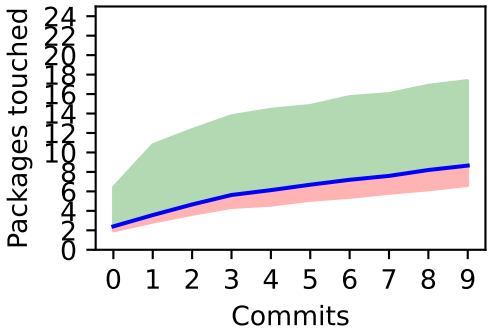
(a) Moderate founder (n=41, μ=8.02, median=4.0 & std=5.85)



(b) Moderate later joiner (n=646, μ=9.64, median=5.0 & std=6.16)

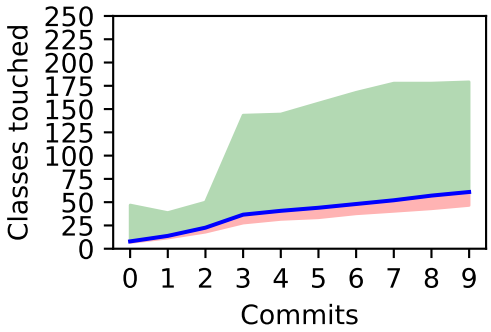


(c) Sustained founder (n=34, μ=6.62, median=3.0 & std=4.25)

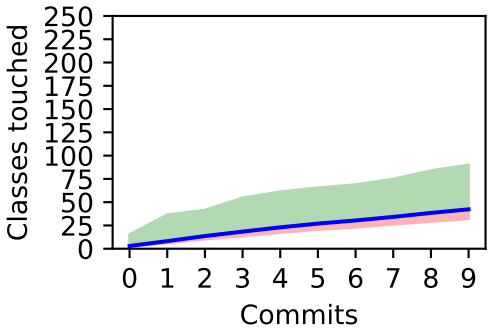


(d) Sustained later joiner (n=203, μ=8.65, median=4.0 & std=6.6)

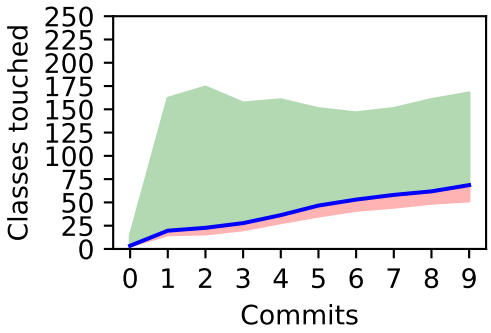
Figure 14: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



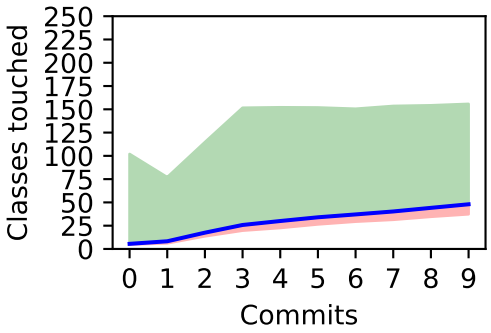
(a) Moderate founder (n=41, μ=60.93, median=19.0 & std=69.12)



(b) Moderate later joiner (n=646, μ=42.32, median=14.0 & std=31.27)

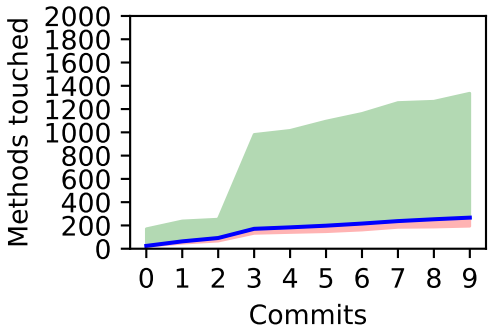


(c) Sustained founder (n=34, μ=68.59, median=15.0 & std=73.09)

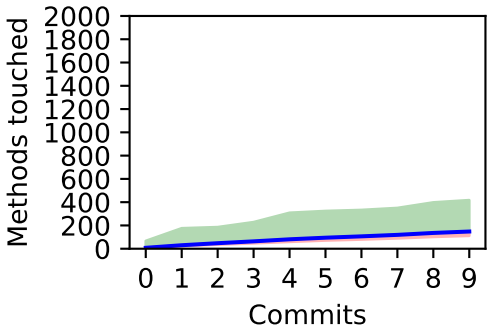


(d) Sustained later joiner (n=203, μ=47.98, median=13.0 & std=62.35)

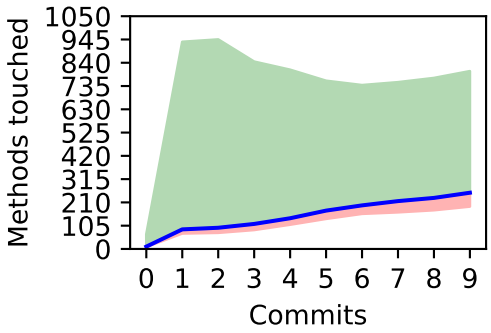
Figure 15: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



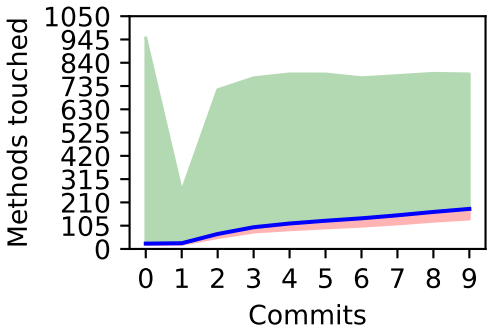
(a) Moderate founder (n=41, μ=267.12, median=59.0 & std=460.61)



(b) Moderate later joiner (n=646, μ=147.78, median=38.0 & std=146.66)



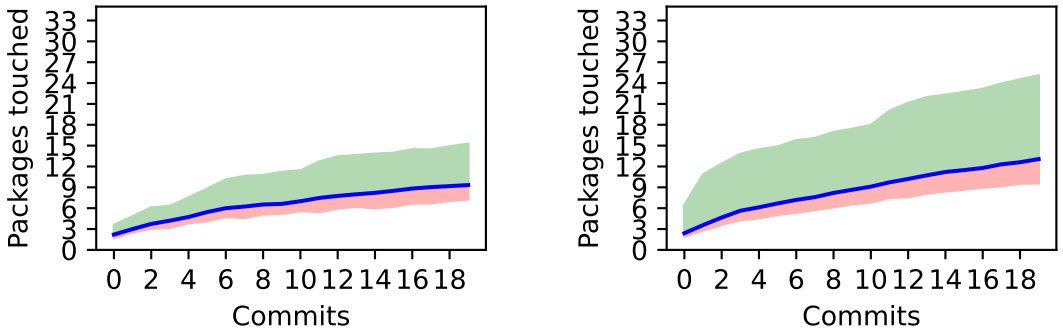
(c) Sustained founder (n=34, μ=253.65, median=42.5 & std=351.3)



(d) Sustained later joiner (n=203, μ=180.7, median=34.0 & std=313.08)

Figure 16: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

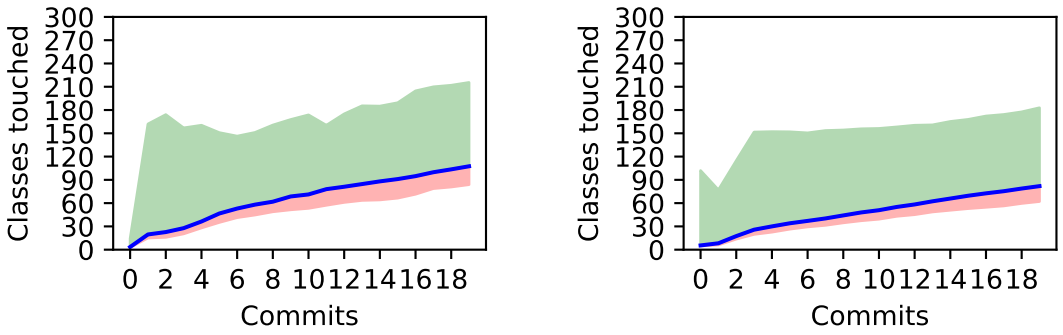
5.3 Time series components touched - For packages touched for first twenty commits



(a) Sustained founder (n=34, μ=9.32, median=4.0 & std=6.14)

(b) Sustained later joiner (n=203, μ=13.06, median=6.0 & std=9.14)

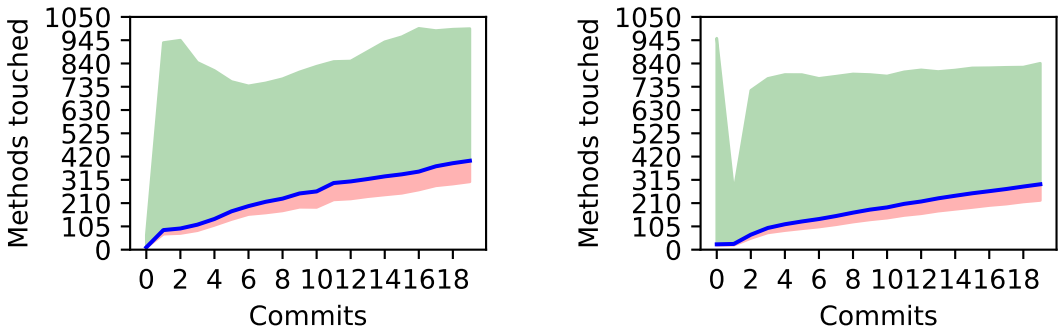
Figure 17: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=107.5, median=30.0 & std=91.26)

(b) Sustained later joiner (n=203, μ=81.91, median=27.0 & std=72.01)

Figure 18: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

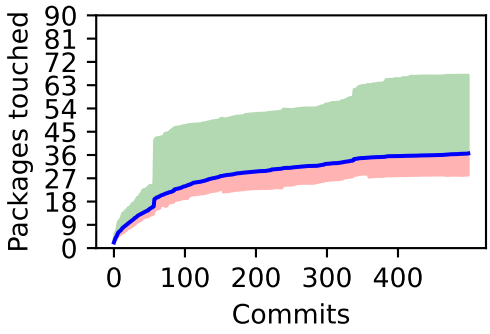


(a) Sustained founder (n=34, μ=401.24, median=85.5 & std=426.09)

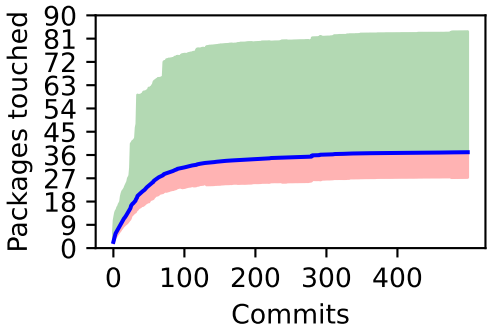
(b) Sustained later joiner (n=203, μ=295.1, median=79.0 & std=349.63)

Figure 19: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

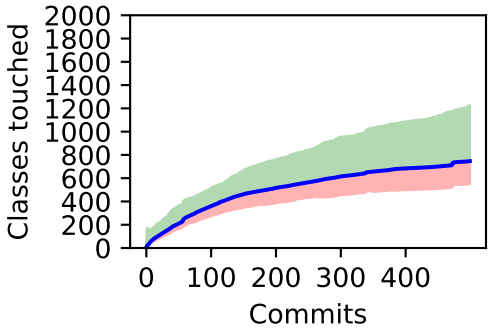
5.4 Time series components touched - For packages touched for commit



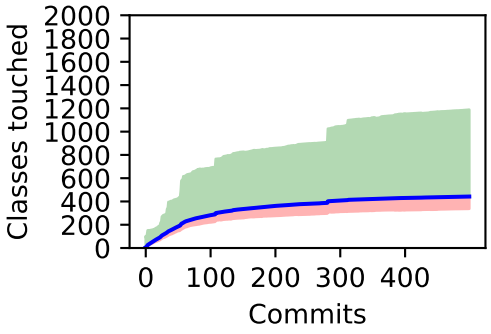
(a) Packages for All founders (n=34, μ=36.62, median=18.0 & std=28.13)



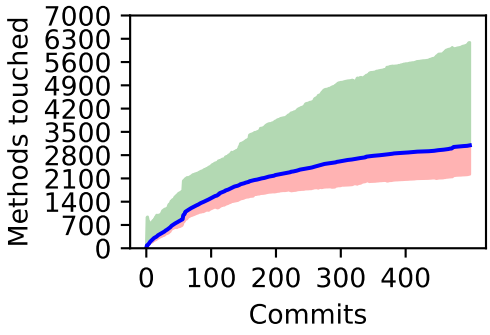
(b) Packages for All joiners (n=203, μ=37.08, median=22.0 & std=36.82)



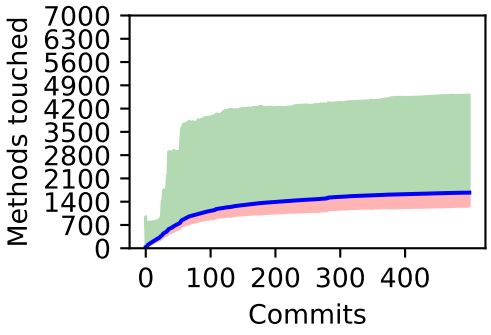
(c) Classes for All founders (n=34, μ=746.59, median=426.0 & std=403.07)



(d) Classes for All joiners (n=203, μ=442.33, median=239.0 & std=426.23)

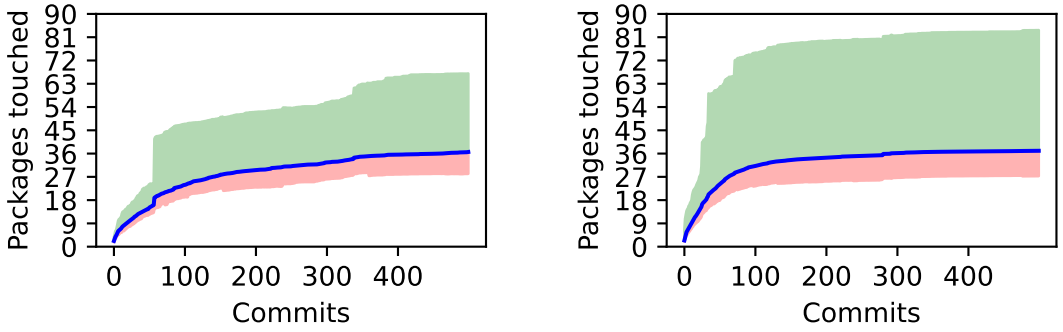


(e) Methods for All founders (n=34, μ=3095.18, median=1645.5 & std=2043.47)



(f) Methods for All joiners (n=203, μ=1672.84, median=838.0 & std=1909.9)

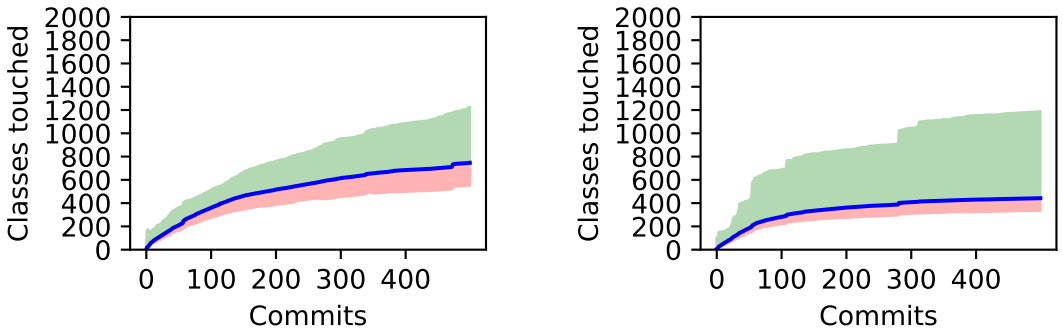
Figure 20: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=36.62, median=18.0 & std=28.13)

(b) Sustained later joiner (n=203, μ=37.08, median=22.0 & std=36.82)

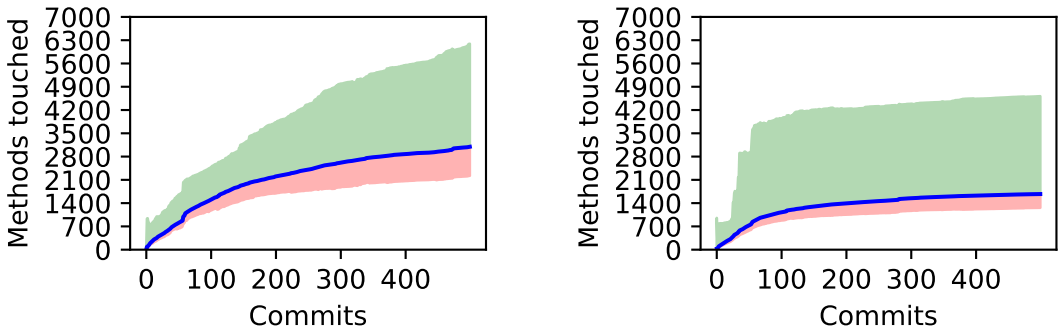
Figure 21: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=746.59, median=426.0 & std=403.07)

(b) Sustained later joiner (n=203, μ=442.33, median=239.0 & std=426.23)

Figure 22: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

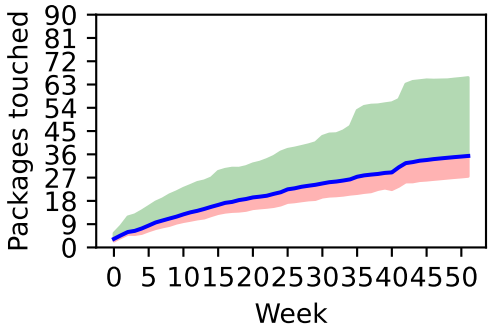


(a) Sustained founder (n=34, μ=3095.18, median=1645.5 & std=2043.47)

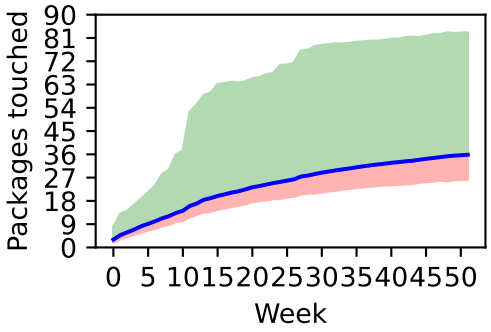
(b) Sustained later joiner (n=203, μ=1672.84, median=838.0 & std=1909.9)

Figure 23: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

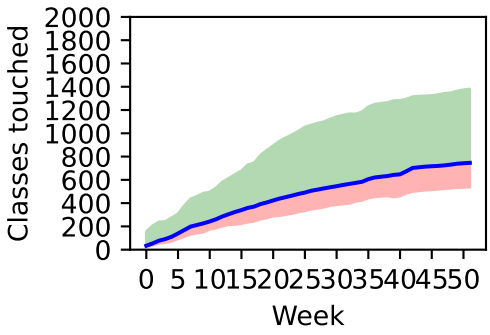
5.5 Time series components touched - For packages touched for each week commit



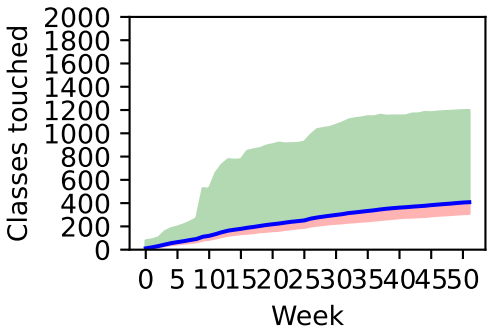
(a) Packages for All founders (n=34, μ=35.38, median=14.0 & std=23.11)



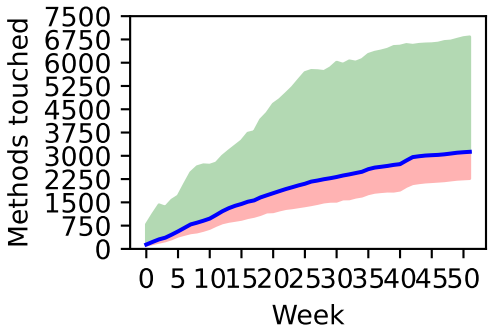
(b) Packages for All joiners (n=203, μ=35.84, median=13.0 & std=33.4)



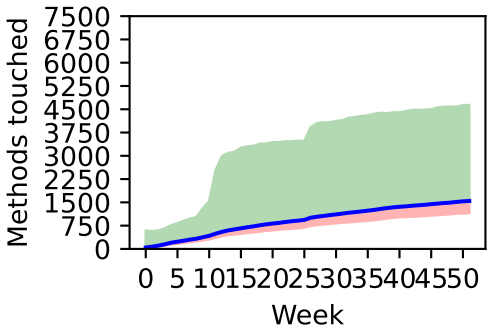
(c) Classes for All founders (n=34, μ=746.15, median=321.28 & std=515.17)



(d) Classes for All joiners (n=203, μ=408.05, median=125.0 & std=440.63)

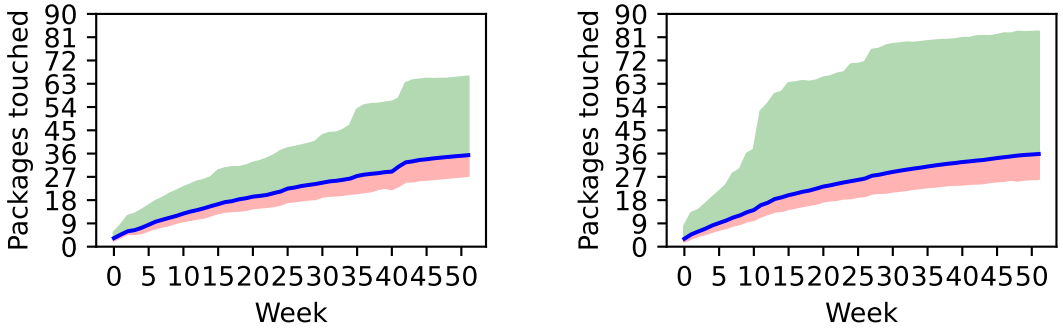


(e) Methods for All founders (n=34, μ=3129.0, median=1296.71 & std=2539.1)



(f) Methods for All joiners (n=203, μ=1545.61, median=435.2 & std=1709.56)

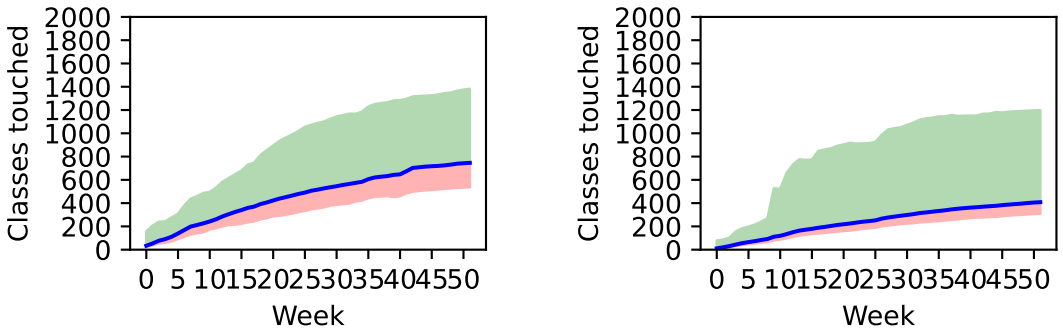
Figure 24: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=35.38, median=14.0 & std=23.11)

(b) Sustained later joiner (n=203, μ=35.84, median=13.0 & std=33.4)

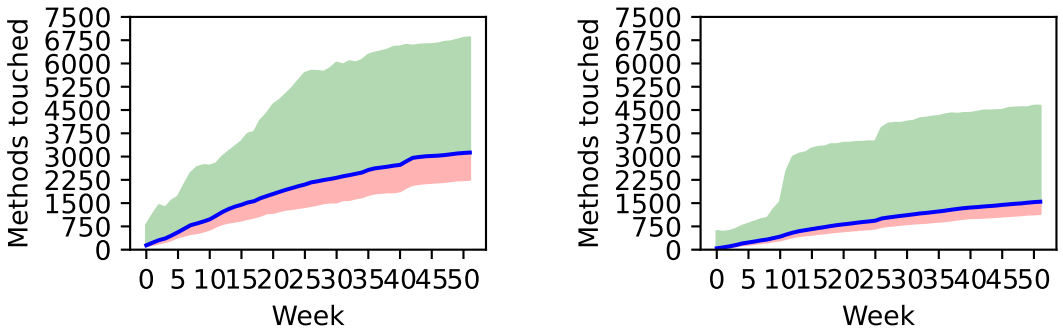
Figure 25: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=746.15, median=321.28 & std=515.17)

(b) Sustained later joiner (n=203, μ=408.05, median=125.0 & std=440.63)

Figure 26: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=3129.0, median=1296.71 & std=2539.1)

(b) Sustained later joiner (n=203, μ=1545.61, median=435.2 & std=1709.56)

Figure 27: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

6 Normality and homoscedasticity - For all components touched for all commits

Certain statistical methods require normality of each sample or the equality of variance. A Shapiro-Wilk Test evaluates whether a sample distribution significantly differs from the normal distribution. A Levene's test evaluates if the independent samples have equal variance.

Table 2: Diagnostic P-Value Thresholds Required for Statistical Method Selection

Statistical Method	**Shapiro-Wilk A (P)**	**Shapiro-Wilk B (P)**	**Levene's Test (P)**
Student's t-test	≥ 0.05 (Normal)	≥ 0.05 (Normal)	≥ 0.05 (Equal Variance)
Standard ANOVA	≥ 0.05 (Normal)	≥ 0.05 (Normal)	≥ 0.05 (Equal Variance)
Welch's t-test	≥ 0.05 (Normal)	≥ 0.05 (Normal)	< 0.05 (Unequal Variance)
Mann-Whitney U test	< 0.05 (Skewed)	< 0.05 (Skewed)	≥ 0.05 (Equal Variance)
Brunner-Münzel test	< 0.05 (Skewed)	< 0.05 (Skewed)	< 0.05 (Unequal Variance)
ART Anova test	< 0.05 (Skewed)	< 0.05 (Skewed)	< 0.05 (Unequal Variance)

Columns below:

- **Component** - Packages, classes or methods touched.
- **First *n* commits** - A sample of the first x number of commits for developers.
- **Contributor Category** Transient, Moderate or Sustained. "vs." equals versus.
- **Sample A** μ - The mean average of the first sample A.
- **Sample A NUmber** - The number of developers the first sample A.
- **Sample B** μ - The mean average of the second sample B.
- **Sample B Number** - The number of developers the first sample B.
- **Percentage Difference in** μ - The percentage difference in means between samples.
- **Sample A Normality** P - Shapiro normality P value of sample A.
- **Sample B Normality** P - Shapiro normality P value of sample B.
- **Variance Equality** P - The Levene variance equality p-value between sample A and B.
- **Statistical Model** - A recommended statistical model of Welch t test; Mann-Whitney; Brunner Münzel
- **Model** P - The P value of the recommended test Welch t test; Mann-Whitney; Brunner Münzel

Table 3: A Shapiro-Wilk test of different categories of founder versus different categories of late joiner sample normality and a Levene's variance test between samples

Component	**First commits** n	**Contributory Category**	**Founder**		**Joiner**		**Percentage Difference in** μ	**Founder Normality** P	**Joiner Normality** P	**Variance Equality** P	**Statistical Models**	**Model** P
			μ	**number**	μ	**number**						
Packages	1	All	1.52	281	1.22	16153	80.37	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	1	Moderate	2.59	41	2.59	646	100.35	<0.001	<0.001	0.8110	Mann-Whitney	0.7564
Packages	1	Sustained	2.21	34	2.40	203	108.98	<0.001	<0.001	0.6501	Mann-Whitney	0.8805
Packages	5	Transient with five or more; moderate & sustained	22.65	84	15.94	1278	70.35	<0.001	<0.001	0.0318	Brunner-Münzel	0.0054
Packages	5	Moderate	15.00	41	15.54	646	103.58	<0.001	<0.001	0.9528	Mann-Whitney	0.6067
Packages	5	Sustained	36.62	34	37.08	203	101.26	<0.001	<0.001	0.9187	Mann-Whitney	0.9397
Packages	10	Moderate & sustained	7.39	75	9.40	849	127.28	<0.001	<0.001	0.3972	Mann-Whitney	0.0031
Packages	10	Moderate	8.02	41	9.64	646	120.11	<0.001	<0.001	0.7772	Mann-Whitney	0.0394
Packages	10	Sustained	6.62	34	8.65	203	130.71	<0.001	<0.001	0.4753	Mann-Whitney	0.1076
Packages	20	Sustained	9.32	34	13.06	203	140.12	<0.001	<0.001	0.3136	Mann-Whitney	0.0435
Classes	1	All	2.75	281	1.27	16153	46.18	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	1	Moderate	7.88	41	2.95	646	37.41	<0.001	<0.001	<0.001	Brunner-Münzel	0.4372
Classes	1	Sustained	3.44	34	5.49	203	159.47	<0.001	<0.001	0.7146	Mann-Whitney	0.1429
Classes	5	Transient with five or more; moderate & sustained	374.23	84	120.59	1278	32.22	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	5	Moderate	143.59	41	82.88	646	57.72	<0.001	<0.001	<0.001	Brunner-Münzel	0.0324
Classes	5	Sustained	746.59	34	442.33	203	59.25	0.0035	<0.001	0.1161	Mann-Whitney	0.0001
Classes	10	Moderate & sustained	64.40	75	43.67	849	67.82	<0.001	<0.001	0.0012	Brunner-Münzel	0.0798
Classes	10	Moderate	60.93	41	42.32	646	69.46	<0.001	<0.001	0.0092	Brunner-Münzel	0.1571
Classes	10	Sustained	68.59	34	47.98	203	69.95	<0.001	<0.001	0.1446	Mann-Whitney	0.2478
Classes	20	Sustained	107.50	34	81.91	203	76.20	<0.001	<0.001	0.0906	Mann-Whitney	0.2857
Methods	1	All	6.90	281	2.00	16153	28.97	<0.001	<0.001	0.0044	Brunner-Münzel	0.0000
Methods	1	Moderate	24.27	41	7.92	646	32.65	<0.001	<0.001	0.0026	Brunner-Münzel	0.5050
Methods	1	Sustained	10.59	34	24.08	203	227.41	<0.001	<0.001	0.7393	Mann-Whitney	0.2351
Methods	5	Transient with five or more; moderate & sustained	1554.73	84	443.83	1278	28.55	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	5	Moderate	605.73	41	297.26	646	49.07	<0.001	<0.001	<0.001	Brunner-Münzel	0.0061
Methods	5	Sustained	3095.18	34	1672.84	203	54.05	<0.001	<0.001	0.0871	Mann-Whitney	0.0000
Methods	10	Moderate & sustained	261.01	75	155.65	849	59.63	<0.001	<0.001	0.0020	Brunner-Münzel	0.0597
Methods	10	Moderate	267.12	41	147.78	646	55.32	<0.001	<0.001	0.0023	Brunner-Münzel	0.1446
Methods	10	Sustained	253.65	34	180.70	203	71.24	<0.001	<0.001	0.3267	Mann-Whitney	0.2291
Methods	20	Sustained	401.24	34	295.10	203	73.55	<0.001	<0.001	0.1120	Mann-Whitney	0.5131

Table 4: A Shapiro-Wilk test of sample A category versus sample B category sample normality and a Levene's variance test between samples

Component	First n commits	Contributory Category	Sample				Percentage Difference in μ	Sample		Variance Equality P	Statistical Models	Model P
			A μ	A number	B μ	B number		A Normality P	B Normality P			
Packages	1	Transient vs. moderate	1.15	15510	2.59	687	225.90	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	1	Transient vs. sustained	1.15	15510	2.38	237	206.89	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	1	Transient vs. moderate & sustained	1.15	15510	2.54	924	221.03	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	1	Moderate vs. sustained	2.59	687	2.38	237	91.58	<0.001	<0.001	0.5986	Mann-Whitney	0.1002
Packages	5	Moderate vs. sustained	15.51	687	37.01	237	238.71	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	10	Moderate vs. sustained	9.54	687	8.36	237	87.60	<0.001	<0.001	0.4602	Mann-Whitney	0.0010
Classes	1	Transient vs. moderate	1.15	15510	3.24	687	281.92	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	1	Transient vs. sustained	1.15	15510	5.19	237	451.72	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	1	Transient vs. moderate & sustained	1.15	15510	3.74	924	325.47	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	1	Moderate vs. sustained	3.24	687	5.19	237	160.23	<0.001	<0.001	0.1255	Mann-Whitney	0.2576
Classes	5	Moderate vs. sustained	86.51	687	485.98	237	561.79	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	10	Moderate vs. sustained	43.43	687	50.94	237	117.28	<0.001	<0.001	0.0223	Brunner-Münzel	0.6723
Methods	1	Transient vs. moderate	1.47	15510	8.90	687	603.66	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	1	Transient vs. sustained	1.47	15510	22.14	237	1501.99	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	1	Transient vs. moderate & sustained	1.47	15510	12.30	924	834.07	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	1	Moderate vs. sustained	8.90	687	22.14	237	248.82	<0.001	<0.001	0.1237	Mann-Whitney	0.1344
Methods	5	Moderate vs. sustained	315.67	687	1876.89	237	594.58	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	10	Moderate vs. sustained	154.90	687	191.16	237	123.41	<0.001	<0.001	0.0503	Mann-Whitney	0.9692

7 Statistical significance test - For all components touched for all commits

A Welch t-test was employed to determine if the mean contributions of Group A differ significantly from Group B. This test is appropriate for continuous data and is robust to unequal variances.

Complementarily, the Mann-Whitney U (MWU) test was used to assess stochastic dominance by ranking data points. This non-parametric approach does not require a normal distribution and is highly resistant to outliers.

The Brunner-Münzel test was employed as a robust non-parametric method to determine if the relative treatment effects of Group A differ significantly from Group B. This test is highly appropriate for skewed count data and is completely robust to unequal variances and large group size imbalances. Complementarily, Cliff's δ and the Probability of Superiority (PS) are provided as robust non-parametric effect sizes to quantify the directional dominance magnitude between the samples.

Columns below:

- **Component** - Packages, classes or method touched.
- **First *n* commits** - A sample of the first x number of commits for developers.
- **Contributor Category** Transient, Moderate or Sustained. "vs." equals versus.
- **Sample A** μ - The mean average of the first sample A. In brackets the number of developers for sample A.
- **Sample B** μ - The mean average of the second sample B. In brackets the number of developers for sample b.
- **Welch Statistic** - This is the magnitude of the change, greater than 2.0 generally indicates a significant change.
- **Welch** P - A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **MWU** P - The Mann-Whitney p-value. A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **BM** P - The Brunner-Münzel asymptotic p-value. A $p < 0.05$ indicates a significant difference in relative effects.
- **Brunner-Münzel W** - The robust test statistic. Large absolute values indicate a significant structural shift.
- **Cliff's** δ - Non-parametric effect size bound between -1 and +1, evaluating directional dominance.
- **Prob. of Superiority (PS)** - The empirical probability (0 to 1) that a randomly selected developer from Sample A outranked a developer from Sample B.

Table 5: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of component (package, class, method) after 1, 5, 10 and 20 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First n commits	Contributor Category	Founder		Joiner		Percentage Difference in μ	Welch		MWU P	BM P	Brunner-Münzel W	Cliff's δ	Prob. Sup. (PS)
			μ	Number	μ	Number		Statistic	P					
Packages	1	All	1.52	281	1.22	16153	80.37	3.61	0.0004	0.0000	0.0000	-5.14	0.129	0.564
Packages	1	Moderate	2.59	41	2.59	646	100.35	-0.02	0.9817	0.7564	0.7537	-0.32	0.028	0.514
Packages	1	Sustained	2.21	34	2.40	203	108.98	-0.55	0.5833	0.8805	0.8804	0.15	-0.015	0.492
Packages	5	Transient with five or more; moderate & sustained	22.65	84	15.94	1278	70.35	2.33	0.0219	0.0041	0.0054	-2.85	0.187	0.593
Packages	5	Moderate	15.00	41	15.54	646	103.58	-0.24	0.8080	0.6067	0.6227	0.50	-0.048	0.476
Packages	5	Sustained	36.62	34	37.08	203	101.26	-0.07	0.9420	0.9397	0.9395	-0.08	0.008	0.504
Packages	10	Moderate & sustained	7.39	75	9.40	849	127.28	-2.42	0.0177	0.0031	0.0046	2.91	-0.206	0.397
Packages	10	Moderate	8.02	41	9.64	646	120.11	-1.29	0.2053	0.0394	0.0504	2.01	-0.191	0.404
Packages	10	Sustained	6.62	34	8.65	203	130.71	-1.85	0.0694	0.1076	0.1319	1.54	-0.172	0.414
Packages	20	Sustained	9.32	34	13.06	203	140.12	-2.31	0.0243	0.0435	0.0572	1.96	-0.216	0.392
Classes	1	All	2.75	281	1.27	16153	46.18	2.62	0.0094	0.0000	0.0000	-5.08	0.108	0.554
Classes	1	Moderate	7.88	41	2.95	646	37.41	1.43	0.1605	0.3900	0.4372	-0.78	0.063	0.531
Classes	1	Sustained	3.44	34	5.49	203	159.47	-0.80	0.4241	0.1429	0.1814	-1.36	0.117	0.559
Classes	5	Transient with five or more; moderate & sustained	374.23	84	120.59	1278	32.22	4.70	0.0000	0.0000	0.0000	-6.46	0.416	0.708
Classes	5	Moderate	143.59	41	82.88	646	57.72	1.97	0.0557	0.0180	0.0324	-2.21	0.220	0.610
Classes	5	Sustained	746.59	34	442.33	203	59.25	2.95	0.0051	0.0001	0.0001	-4.36	0.434	0.717
Classes	10	Moderate & sustained	64.40	75	43.67	849	67.82	2.04	0.0446	0.0557	0.0798	-1.77	0.133	0.567
Classes	10	Moderate	60.93	41	42.32	646	69.46	1.36	0.1807	0.1158	0.1571	-1.44	0.146	0.573
Classes	10	Sustained	68.59	34	47.98	203	69.95	1.30	0.2003	0.2478	0.2786	-1.10	0.124	0.562
Classes	20	Sustained	107.50	34	81.91	203	76.20	1.34	0.1863	0.2857	0.3224	-1.00	0.115	0.557
Methods	1	All	6.90	281	2.00	16153	28.97	2.35	0.0193	0.0000	0.0000	-5.30	0.117	0.558
Methods	1	Moderate	24.27	41	7.92	646	32.65	1.30	0.2026	0.4647	0.5050	-0.67	0.056	0.528
Methods	1	Sustained	10.59	34	24.08	203	227.41	-0.78	0.4382	0.2351	0.2715	-1.11	0.098	0.549
Methods	5	Transient with five or more; moderate & sustained	1554.73	84	443.83	1278	28.55	4.45	0.0000	0.0000	0.0000	-7.01	0.442	0.721
Methods	5	Moderate	605.73	41	297.26	646	49.07	2.01	0.0511	0.0027	0.0061	-2.88	0.279	0.640
Methods	5	Sustained	3095.18	34	1672.84	203	54.05	2.86	0.0067	0.0000	0.0000	-4.69	0.458	0.729
Methods	10	Moderate & sustained	261.01	75	155.65	849	59.63	1.73	0.0884	0.0384	0.0597	-1.91	0.144	0.572
Methods	10	Moderate	267.12	41	147.78	646	55.32	1.24	0.2205	0.1041	0.1446	-1.49	0.151	0.576
Methods	10	Sustained	253.65	34	180.70	203	71.24	1.00	0.3240	0.2291	0.2666	-1.13	0.129	0.565
Methods	20	Sustained	401.24	34	295.10	203	73.55	1.16	0.2533	0.5131	0.5561	-0.59	0.070	0.535

Table 6: Statistical significance tests of sample A category versus sample B category touches of different granularities of component (package, class, method) after 1, 5 and 10 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First n commits	Contributor Category	Sample				Percentage Difference in μ	Welch		MWU P	BM P	Brunner-Münzel W	Cliff's δ	Prob. Sup. (PS)
			A μ	A Number	B μ	B Number		Statistic	P					
Packages	1	Transient vs. moderate	1.15	15510	2.59	687	225.90	-12.35	0.0000	0.0000	0.0000	27.59	-0.526	0.237
Packages	1	Transient vs. sustained	1.15	15510	2.38	237	206.89	-7.23	0.0000	0.0000	0.0000	14.28	-0.468	0.266
Packages	1	Transient vs. moderate & sustained	1.15	15510	2.54	924	221.03	-14.27	0.0000	0.0000	0.0000	30.91	-0.511	0.244
Packages	1	Moderate vs. sustained	2.59	687	2.38	237	91.58	1.06	0.2896	0.1002	0.0995	-1.65	0.068	0.534
Packages	5	Moderate vs. sustained	15.51	687	37.01	237	238.71	-8.40	0.0000	0.0000	0.0000	11.65	-0.463	0.268
Packages	10	Moderate vs. sustained	9.54	687	8.36	237	87.60	2.06	0.0397	0.0010	0.0012	-3.27	0.142	0.571
Classes	1	Transient vs. moderate	1.15	15510	3.24	687	281.92	-6.44	0.0000	0.0000	0.0000	14.31	-0.245	0.378
Classes	1	Transient vs. sustained	1.15	15510	5.19	237	451.72	-2.07	0.0397	0.0000	0.0000	7.55	-0.210	0.395
Classes	1	Transient vs. moderate & sustained	1.15	15510	3.74	924	325.47	-4.66	0.0000	0.0000	0.0000	16.15	-0.236	0.382
Classes	1	Moderate vs. sustained	3.24	687	5.19	237	160.23	-0.99	0.3256	0.2576	0.2451	-1.16	0.038	0.519
Classes	5	Moderate vs. sustained	86.51	687	485.98	237	561.79	-11.13	0.0000	0.0000	0.0000	42.42	-0.834	0.083
Classes	10	Moderate vs. sustained	43.43	687	50.94	237	117.28	-1.48	0.1409	0.6684	0.6723	-0.42	0.019	0.509
Methods	1	Transient vs. moderate	1.47	15510	8.90	687	603.66	-5.75	0.0000	0.0000	0.0000	15.70	-0.279	0.360
Methods	1	Transient vs. sustained	1.47	15510	22.14	237	1501.99	-1.46	0.1458	0.0000	0.0000	7.95	-0.228	0.386
Methods	1	Transient vs. moderate & sustained	1.47	15510	12.30	924	834.07	-2.88	0.0041	0.0000	0.0000	17.55	-0.266	0.367
Methods	1	Moderate vs. sustained	8.90	687	22.14	237	248.82	-0.93	0.3527	0.1344	0.1240	-1.54	0.053	0.526
Methods	5	Moderate vs. sustained	315.67	687	1876.89	237	594.58	-10.46	0.0000	0.0000	0.0000	41.88	-0.823	0.088
Methods	10	Moderate vs. sustained	154.90	687	191.16	237	123.41	-1.44	0.1497	0.9692	0.9696	0.04	-0.002	0.499

Statistical Models

Welch's t-test To specifically compare the two primary roles (**Sample A** vs. **Sample B**) without assuming equal variances or sample sizes, Welch's t-test is employed. The t-statistic is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \tag{1}$$

Where:

- $\bar{X}_1, \bar{X}_2$ are the sample means of Sample A and Sample B.
- s_1^2, s_2^2 are the sample variances.
- N_1, N_2 are the sample sizes.

The degrees of freedom (ν) for this test are calculated using the Welch-Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1 - 1} + \frac{(s_2^2/N_2)^2}{N_2 - 1}} \tag{2}$$

Mann-Whitney U Test To evaluate the differences in the stochastic dominance of architectural component exploration between primary roles (**Sample A** vs. **Sample B**) without assuming a normal distribution, the non-parametric Mann-Whitney U test (Wilcoxon rank-sum test) is employed.

The data from both independent groups are combined and ranked globally from lowest (1) to highest ($N_1 + N_2$), resolving ties by assigning the average rank. The U statistic for Sample A (U_1) is calculated as:

$$U_1 = R_1 - \frac{N_1(N_1 + 1)}{2} \tag{3}$$

Where:

- N_1 is the sample size of Sample A.
- R_1 is the sum of the ranks assigned to Sample A.
- The total number of pairwise comparisons (maximum possible pairs) is defined by $N_1 \times N_2$.

Normal Approximation for Large Samples Given the substantial sample sizes present in repository participant metrics, the distribution of the U statistic rapidly approaches a normal distribution. The asymptotic significance is determined using a standard normal Z-score approximation:

$$Z = \frac{U_1 - \mu_U}{\sigma_U} \tag{4}$$

Where the expected mean (μ_U) and standard deviation (σ_U) under the null hypothesis of identical distributions are defined as:

$$\mu_U = \frac{N_1 N_2}{2}, \quad \sigma_U = \sqrt{\frac{N_1 N_2 (N_1 + N_2 + 1)}{12}} \tag{5}$$

A calculated p-value lower than the alpha threshold ($p < 0.05$) indicates a statistically significant difference in the rank-order distributions of the two groups, demonstrating that one developer role systematically outranks the other in codebase exploration.

Brunner-Münzel Test To test for differences in the stochastic distributions of primary roles (**Sample A** vs. **Sample B**) without assuming normal distribution or variance homogeneity, the Brunner-Münzel test is applied. The test statistic W is defined as:

$$W = \sqrt{\frac{N_1 N_2}{N_1 + N_2}} \cdot \frac{\hat{p} - \frac{1}{2}}{\sqrt{\hat{\sigma}^2}} \tag{6}$$

Where $\hat{p}$ represents the relative treatment effect (Probability of Superiority), reflecting the probability that a random observation from Sample A is larger than one from Sample B plus half the probability that they are equal.

Cliff's δ Effect Size To complement hypothesis significance testing with a distribution-free magnitude metric, Cliff's δ is tracked to express the degree of dominance of one sample over the other:

$$\delta = \frac{\sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \text{sign}(X_{1i} - X_{2j})}{N_1 N_2} \tag{7}$$

Where the sign function yields $+1$ when an element of Sample A is larger than Sample B, 0 if equal, and -1 if smaller. Effect sizes are interpreted using standard empirical thresholds ($|\delta| < 0.147$ is negligible, ≥ 0.147 is small, ≥ 0.330 is medium, and ≥ 0.474 is large).

8 Generated standard anova - For all components touched for all commits

An Anova table for all components touched on average.

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

Explanation of Table Columns:

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

Explanation of Table Rows:

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Role Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage component exploration.
- **Category(M/S):** Tests the *Main Category Effect* of the eventual contributor catregory (Transient vs Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the "noise" or variance within the groups that cannot be explained by Role or Tenure.

Table 7: Anova results for all components, number of commits and categories of developers

Component	First n commits	Number of Developers	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
packages	1	16434	Role (F/LJ)	0.00	1	0.00	0.9606	0.0000
packages	1	16434	Category (T/M/S)	84.40	2	32.56	0.0000	0.0039
packages	1	16434	Interaction	1.54	2	0.60	0.5510	0.0001
packages	1	16434	Residual	21289.91	16428	-	-	
packages	5	1362	Role (F/LJ)	11.12	1	0.03	0.8640	0.0000
packages	5	1362	Category (M/S)	11907.40	2	15.71	0.0000	0.0222
packages	5	1362	Interaction	12.31	2	0.02	0.9839	0.0000
packages	5	1362	Residual	514037.69	1356	-	-	
packages	10	924	Role (F/LJ)	100.35	1	1.79	0.1809	0.0018
packages	10	924	Category (M/S)	36.78	1	0.66	0.4177	0.0007
packages	10	924	Interaction	2.92	1	0.05	0.8195	0.0001
packages	10	924	Residual	51484.41	920	-	-	
classes	1	16434	Role (F/LJ)	937.29	1	47.88	0.0000	0.0029
classes	1	16434	Category (T/M/S)	1359.09	2	34.71	0.0000	0.0042
classes	1	16434	Interaction	915.16	2	23.37	0.0000	0.0028
classes	1	16434	Residual	321604.01	16428	-	-	
classes	5	1362	Role (F/LJ)	142055.48	1	2.54	0.1109	0.0017
classes	5	1362	Category (M/S)	8035870.96	2	71.98	0.0000	0.0935
classes	5	1362	Interaction	1215900.38	2	10.89	0.0000	0.0142
classes	5	1362	Residual	75688676.66	1356	-	-	
classes	10	924	Role (F/LJ)	13347.02	1	4.68	0.0308	0.0048
classes	10	924	Category (M/S)	1090.98	1	0.38	0.5365	0.0004
classes	10	924	Interaction	66.46	1	0.02	0.8787	0.0000
classes	10	924	Residual	2625443.61	920	-	-	
methods	1	16434	Role (F/LJ)	10298.74	1	12.69	0.0004	0.0008
methods	1	16434	Category (T/M/S)	16234.54	2	10.01	0.0000	0.0012
methods	1	16434	Interaction	14996.83	2	9.24	0.0001	0.0011
methods	1	16434	Residual	13328279.01	16428	-	-	
methods	5	1362	Role (F/LJ)	3668624.09	1	3.78	0.0519	0.0025
methods	5	1362	Category (M/S)	137755503.49	2	71.05	0.0000	0.0920
methods	5	1362	Interaction	25729945.47	2	13.27	0.0000	0.0172
methods	5	1362	Residual	1314541598.99	1356	-	-	
methods	10	924	Role (F/LJ)	549077.04	1	7.22	0.0073	0.0075
methods	10	924	Category (M/S)	3374.83	1	0.04	0.8332	0.0000
methods	10	924	Interaction	35707.01	1	0.47	0.4934	0.0005
methods	10	924	Residual	69978889.05	920	-	-	

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{8}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{9}$$

9 Tukey hsd - For all components touched for all commits

A Tukey Honestly Significant Difference table for all components touched on average.

Post-hoc Analysis (Tukey HSD): Following the ANOVA, a Tukey Honestly Significant Difference (HSD) test was performed. While ANOVA identifies if at least one group differs, Tukey HSD performs pairwise comparisons to identify specific differences between developer roles and tenure categories while controlling for Type I error rates across multiple comparisons.

Table 8: Significant Tukey HSD results for components and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	1	16434	Founder Moderate	Founder Sustained	-4.44	0.0002
classes	1	16434	Founder Moderate	Founder Transient	-6.26	0.0000
classes	1	16434	Founder Moderate	Later Joiner Moderate	-4.93	0.0000
classes	1	16434	Founder Moderate	Later Joiner Sustained	-2.39	0.0200
classes	1	16434	Founder Moderate	Later Joiner Transient	-6.73	0.0000
classes	1	16434	Founder Sustained	Later Joiner Transient	-2.30	0.0300
classes	1	16434	Founder Transient	Later Joiner Moderate	1.33	0.0024
classes	1	16434	Founder Transient	Later Joiner Sustained	3.87	0.0000
classes	1	16434	Later Joiner Moderate	Later Joiner Sustained	2.54	0.0000
classes	1	16434	Later Joiner Moderate	Later Joiner Transient	-1.80	0.0000
classes	1	16434	Later Joiner Sustained	Later Joiner Transient	-4.34	0.0000
classes	5	1362	Founder Moderate	Founder Sustained	603.00	0.0000
classes	5	1362	Founder Moderate	Later Joiner Sustained	298.75	0.0000
classes	5	1362	Founder Moderate	Later Joiner Transient	-118.47	0.0267
classes	5	1362	Founder Sustained	Founder Transient	-728.37	0.0000
classes	5	1362	Founder Sustained	Later Joiner Moderate	-663.70	0.0000
classes	5	1362	Founder Sustained	Later Joiner Sustained	-304.25	0.0000
classes	5	1362	Founder Sustained	Later Joiner Transient	-721.47	0.0000
classes	5	1362	Founder Transient	Later Joiner Sustained	424.11	0.0000
classes	5	1362	Later Joiner Moderate	Later Joiner Sustained	359.45	0.0000
classes	5	1362	Later Joiner Moderate	Later Joiner Transient	-57.77	0.0013
classes	5	1362	Later Joiner Sustained	Later Joiner Transient	-417.22	0.0000
classes	10	924	Founder Sustained	Later Joiner Moderate	-26.27	0.0272
methods	1	16434	Founder Moderate	Founder Transient	-21.43	0.0002
methods	1	16434	Founder Moderate	Later Joiner Moderate	-16.34	0.0049
methods	1	16434	Founder Moderate	Later Joiner Transient	-22.81	0.0000
methods	1	16434	Founder Transient	Later Joiner Sustained	21.24	0.0000
methods	1	16434	Later Joiner Moderate	Later Joiner Sustained	16.15	0.0000
methods	1	16434	Later Joiner Moderate	Later Joiner Transient	-6.47	0.0000
methods	1	16434	Later Joiner Sustained	Later Joiner Transient	-22.62	0.0000
methods	5	1362	Founder Moderate	Founder Sustained	2489.44	0.0000
methods	5	1362	Founder Moderate	Later Joiner Sustained	1067.11	0.0000
methods	5	1362	Founder Moderate	Later Joiner Transient	-522.74	0.0151
methods	5	1362	Founder Sustained	Founder Transient	-3036.73	0.0000
methods	5	1362	Founder Sustained	Later Joiner Moderate	-2797.92	0.0000
methods	5	1362	Founder Sustained	Later Joiner Sustained	-1422.33	0.0000
methods	5	1362	Founder Sustained	Later Joiner Transient	-3012.19	0.0000
methods	5	1362	Founder Transient	Later Joiner Sustained	1614.40	0.0000
methods	5	1362	Later Joiner Moderate	Later Joiner Sustained	1375.59	0.0000
methods	5	1362	Later Joiner Moderate	Later Joiner Transient	-214.26	0.0065
methods	5	1362	Later Joiner Sustained	Later Joiner Transient	-1589.85	0.0000
methods	10	924	Founder Moderate	Later Joiner Moderate	-119.34	0.0368

Table 9: Not Significant Tukey HSD results for components and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	1	16434	Founder Sustained	Founder Transient	-1.82	0.2249
classes	1	16434	Founder Sustained	Later Joiner Moderate	-0.49	0.9885
classes	1	16434	Founder Sustained	Later Joiner Sustained	2.05	0.1252
classes	1	16434	Founder Transient	Later Joiner Transient	-0.47	0.6487
classes	5	1362	Founder Moderate	Founder Transient	-125.36	0.7015
classes	5	1362	Founder Moderate	Later Joiner Moderate	-60.70	0.6018
classes	5	1362	Founder Transient	Later Joiner Moderate	64.66	0.9648
classes	5	1362	Founder Transient	Later Joiner Transient	6.89	1.0000
classes	10	924	Founder Moderate	Founder Sustained	7.66	0.9262
classes	10	924	Founder Moderate	Later Joiner Moderate	-18.61	0.1346
classes	10	924	Founder Moderate	Later Joiner Sustained	-12.95	0.4899
classes	10	924	Founder Sustained	Later Joiner Sustained	-20.61	0.1598
classes	10	924	Later Joiner Moderate	Later Joiner Sustained	5.66	0.5524
classes	20	237	Founder Sustained	Later Joiner Sustained	-25.59	0.1034
methods	1	16434	Founder Moderate	Founder Sustained	-13.68	0.3028
methods	1	16434	Founder Moderate	Later Joiner Sustained	-0.19	1.0000
methods	1	16434	Founder Sustained	Founder Transient	-7.75	0.6832
methods	1	16434	Founder Sustained	Later Joiner Moderate	-2.66	0.9949
methods	1	16434	Founder Sustained	Later Joiner Sustained	13.49	0.1084
methods	1	16434	Founder Sustained	Later Joiner Transient	-9.13	0.4224
methods	1	16434	Founder Transient	Later Joiner Moderate	5.09	0.2226
methods	1	16434	Founder Transient	Later Joiner Transient	-1.38	0.9831
methods	5	1362	Founder Moderate	Founder Transient	-547.29	0.6579
methods	5	1362	Founder Moderate	Later Joiner Moderate	-308.48	0.3749
methods	5	1362	Founder Transient	Later Joiner Moderate	238.81	0.9792
methods	5	1362	Founder Transient	Later Joiner Transient	24.55	1.0000
methods	10	924	Founder Moderate	Founder Sustained	-13.47	0.9967
methods	10	924	Founder Moderate	Later Joiner Sustained	-86.42	0.2598
methods	10	924	Founder Sustained	Later Joiner Moderate	-105.87	0.1291
methods	10	924	Founder Sustained	Later Joiner Sustained	-72.95	0.4825
methods	10	924	Later Joiner Moderate	Later Joiner Sustained	32.92	0.4480
methods	20	237	Founder Sustained	Later Joiner Sustained	-106.14	0.1554

Post-hoc Analysis: Tukey HSD

Following the Two-Way ANOVA, a Tukey Honestly Significant Difference (HSD) test is used to conduct pairwise comparisons. This test determines which specific group means differ significantly while maintaining a constant family-wise error rate.

The studentized range statistic (q) for comparing two group means is defined as:

$$q = \frac{\bar{Y}_{max} - \bar{Y}_{min}}{\sqrt{\frac{MS_{residual}}{n}}} \tag{10}$$

Where:

- $\bar{Y}_{max}$ and $\bar{Y}_{min}$ are the larger and smaller means of the two groups being compared.
- $MS_{residual}$ is the Mean Square Error derived from the ANOVA residual.
- n is the number of observations in each group (adjusted for unequal sample sizes via the harmonic mean).

A pair of groups is considered significantly different if the absolute difference between their means exceeds the Honestly Significant Difference (HSD):

$$HSD = q_{\alpha,k,df}\sqrt{\frac{MS_{residual}}{n}} \tag{11}$$

Where $q_{\alpha,k,df}$ is the critical value from the studentized range distribution based on the significance level ($\alpha = 0.05$), the number of groups (k), and the degrees of freedom (df) from the ANOVA residual.

10 Generated art anova - For all components touched for all commits

An **Aligned Rank Transform (ART)** Anova table for all components touched on average. The statistical significance of architectural exploration is evaluated using a dual-framework modeling methodology. A standard parametric ordinary least squares (**OLS**) ANOVA is presented alongside a robust non-parametric ANOVA. This side-by-side presentation explicitly demonstrates where extreme right-skew and outlier-induced variance in repository mining data mask true behavioral effects in traditional linear models, which the rank-transformed framework uncovers.

Explanation of Combined Table Columns:

- **Component / Commits (n) / Devs (N):** Define the experimental scope, tracking the architectural tier (packages, classes, or methods), the longitudinal lifecycle snapshot threshold, and the active developer population size.
- **Factor:** The independent categorical variables or interactions being tested for statistical significance.
- **OLS df:** Parametric numerator degrees of freedom based on factor grouping configurations.
- **OLS F-Stat:** The traditional variance ratio metric calculated from raw values.
- **OLS P-Value:** Parametric significance probability. Note how heteroscedastic noise frequently inflates these values toward non-significance ($p > 0.05$).
- **OLS η^2 (Eta-Squared):** The total proportion of raw variance explained exclusively by this standalone factor.
- **ART df (Resid):** Formatted as df_{effect} (df_{residual}). Because the ART methodology conducts independent alignment procedures for each factor before ranking, it assigns the exact denominator error degrees of freedom dynamically alongside its numerator.
- **ART F-Stat:** The robust variance ratio calculated over the aligned, ranked data layers.
- **ART P-Value:** Non-parametric significance probability, providing the authoritative test for heavy-tailed software metrics.
- **ART Part. η^2 (Partial Eta-Squared):** The effect size over ranked data layers, indicating the proportion of variance explained by this specific factor after controlling for alternate main effects.

Explanation of Fixed Table Rows:

- **Role (F/LJ):** Isolates the *Main Role Effect* to evaluate whether the timing of project entry (Founder vs. Later Joiner) significantly alters exploration breadth on its own.
- **Category (T/M/S or M/S):** Isolates the *Main Category Effect* to determine whether a contributor's eventual longevity footprint (Transient, Moderate, or Sustained) dictates early behavior.
- **Interaction:** Evaluates the *Interaction Effect* (Role × Category). A significant interaction indicates that the impact of entry role is not uniform, but behaves completely differently depending on the eventual tenure category of the contributor.

Table 10: ART Anova results for all components, number of commits and categories of developers

Component	First n commits	Number of Developers	Factor	OLS				ART			
				df	F-Stat	P Value	η^2	df (Resid)	F-Stat	P Value	Part. η^2
packages	1	16434	Role (F/LJ)	1	0.00	0.9606	0.0000	1 (16428)	265.55	0.0000	0.0159
packages	1	16434	Category (T/M/S)	2	32.56	0.0000	0.0039	2 (16428)	801.56	0.0000	0.0889
packages	1	16434	Interaction	2	0.60	0.5510	0.0001	2 (16428)	132.65	0.0000	0.0159
packages	5	1362	Role (F/LJ)	1	0.03	0.8640	0.0000	1 (1356)	18.60	0.0000	0.0135
packages	5	1362	Category (M/S)	2	15.71	0.0000	0.0222	2 (1356)	51.18	0.0000	0.0702
packages	5	1362	Interaction	2	0.02	0.9839	0.0000	2 (1356)	2.95	0.0529	0.0043
packages	10	924	Role (F/LJ)	1	1.79	0.1809	0.0018	1 (920)	5.36	0.0209	0.0058
packages	10	924	Category (M/S)	1	0.66	0.4177	0.0007	1 (920)	3.19	0.0744	0.0035
packages	10	924	Interaction	1	0.05	0.8195	0.0001	1 (920)	0.04	0.8387	0.0000
classes	1	16434	Role (F/LJ)	1	47.88	0.0000	0.0029	1 (16428)	463.03	0.0000	0.0274
classes	1	16434	Category (T/M/S)	2	34.71	0.0000	0.0042	2 (16428)	912.14	0.0000	0.0999
classes	1	16434	Interaction	2	23.37	0.0000	0.0028	2 (16428)	305.46	0.0000	0.0359
classes	5	1362	Role (F/LJ)	1	2.54	0.1109	0.0017	1 (1356)	100.03	0.0000	0.0687
classes	5	1362	Category (M/S)	2	71.98	0.0000	0.0935	2 (1356)	150.39	0.0000	0.1815
classes	5	1362	Interaction	2	10.89	0.0000	0.0142	2 (1356)	30.33	0.0000	0.0428
classes	10	924	Role (F/LJ)	1	4.68	0.0308	0.0048	1 (920)	5.70	0.0172	0.0062
classes	10	924	Category (M/S)	1	0.38	0.5365	0.0004	1 (920)	0.11	0.7391	0.0001
classes	10	924	Interaction	1	0.02	0.8787	0.0000	1 (920)	0.16	0.6896	0.0002
methods	1	16434	Role (F/LJ)	1	12.69	0.0004	0.0008	1 (16428)	382.32	0.0000	0.0227
methods	1	16434	Category (T/M/S)	2	10.01	0.0000	0.0012	2 (16428)	801.24	0.0000	0.0889
methods	1	16434	Interaction	2	9.24	0.0001	0.0011	2 (16428)	374.01	0.0000	0.0436
methods	5	1362	Role (F/LJ)	1	3.78	0.0519	0.0025	1 (1356)	102.31	0.0000	0.0702
methods	5	1362	Category (M/S)	2	71.05	0.0000	0.0920	2 (1356)	141.78	0.0000	0.1730
methods	5	1362	Interaction	2	13.27	0.0000	0.0172	2 (1356)	29.99	0.0000	0.0424
methods	10	924	Role (F/LJ)	1	7.22	0.0073	0.0075	1 (920)	9.37	0.0023	0.0101
methods	10	924	Category (M/S)	1	0.04	0.8332	0.0000	1 (920)	1.29	0.2568	0.0014
methods	10	924	Interaction	1	0.47	0.4934	0.0005	1 (920)	0.69	0.4070	0.0007

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{12}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Later Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{13}$$

11 Holm bonferroni - For all components touched for all commits

A Pairwise Post-hoc Analysis with Holm-Bonferroni Correction table for all components touched on average.

Post-hoc Pairwise Analysis (Holm-Bonferroni): Following the multi-factor significance analysis, non-parametric pairwise comparisons were executed to isolate directional stochastic dominance patterns between groups. To counteract family-wise Type I error inflation across multiple group permutations without experiencing the severe statistical power depletion of a traditional Bonferroni adjustment, a sequentially rejective Holm-Bonferroni step-down correction is applied.

Table 11: Significant Holm-Bonferroni Pairwise results for packages and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
packages	1	16434	Founder Moderate	Founder Transient	1.39	0.0000
packages	1	16434	Founder Moderate	Later Joiner Transient	1.44	0.0000
packages	1	16434	Founder Sustained	Founder Transient	1.01	0.0000
packages	1	16434	Founder Sustained	Later Joiner Transient	1.06	0.0000
packages	1	16434	Founder Transient	Later Joiner Moderate	-1.40	0.0000
packages	1	16434	Founder Transient	Later Joiner Sustained	-1.21	0.0000
packages	1	16434	Later Joiner Moderate	Later Joiner Transient	1.45	0.0000
packages	1	16434	Later Joiner Sustained	Later Joiner Transient	1.26	0.0000
packages	5	1362	Founder Moderate	Founder Sustained	-21.62	0.0013
packages	5	1362	Founder Moderate	Founder Transient	10.22	0.0252
packages	5	1362	Founder Moderate	Later Joiner Sustained	-22.08	0.0000
packages	5	1362	Founder Moderate	Later Joiner Transient	8.47	0.0000
packages	5	1362	Founder Sustained	Founder Transient	31.84	0.0004
packages	5	1362	Founder Sustained	Later Joiner Moderate	21.08	0.0000
packages	5	1362	Founder Sustained	Later Joiner Transient	30.08	0.0000
packages	5	1362	Founder Transient	Later Joiner Moderate	-10.76	0.0052
packages	5	1362	Founder Transient	Later Joiner Sustained	-32.30	0.0001
packages	5	1362	Later Joiner Moderate	Later Joiner Sustained	-21.54	0.0000
packages	5	1362	Later Joiner Moderate	Later Joiner Transient	9.00	0.0000
packages	5	1362	Later Joiner Sustained	Later Joiner Transient	30.55	0.0000
packages	10	924	Founder Sustained	Later Joiner Moderate	-3.02	0.0258
packages	10	924	Later Joiner Moderate	Later Joiner Sustained	0.99	0.0258
packages	20	237	Founder Sustained	Later Joiner Sustained	-3.74	0.0435

Table 12: Significant Holm-Bonferroni Pairwise results for classes and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	1	16434	Founder Moderate	Founder Transient	6.26	0.0001
classes	1	16434	Founder Moderate	Later Joiner Transient	6.73	0.0000
classes	1	16434	Founder Sustained	Founder Transient	1.82	0.0001
classes	1	16434	Founder Sustained	Later Joiner Transient	2.30	0.0000
classes	1	16434	Founder Transient	Later Joiner Moderate	-1.33	0.0000
classes	1	16434	Founder Transient	Later Joiner Sustained	-3.87	0.0006
classes	1	16434	Founder Transient	Later Joiner Transient	0.47	0.0001
classes	1	16434	Later Joiner Moderate	Later Joiner Transient	1.80	0.0000
classes	1	16434	Later Joiner Sustained	Later Joiner Transient	4.34	0.0000
classes	5	1362	Founder Moderate	Founder Sustained	-603.00	0.0000
classes	5	1362	Founder Moderate	Founder Transient	125.36	0.0008
classes	5	1362	Founder Moderate	Later Joiner Moderate	60.70	0.0360
classes	5	1362	Founder Moderate	Later Joiner Sustained	-298.75	0.0000
classes	5	1362	Founder Moderate	Later Joiner Transient	118.47	0.0000
classes	5	1362	Founder Sustained	Founder Transient	728.37	0.0000
classes	5	1362	Founder Sustained	Later Joiner Moderate	663.70	0.0000
classes	5	1362	Founder Sustained	Later Joiner Sustained	304.25	0.0003
classes	5	1362	Founder Sustained	Later Joiner Transient	721.47	0.0000
classes	5	1362	Founder Transient	Later Joiner Moderate	-64.66	0.0004
classes	5	1362	Founder Transient	Later Joiner Sustained	-424.11	0.0000
classes	5	1362	Later Joiner Moderate	Later Joiner Sustained	-359.45	0.0000
classes	5	1362	Later Joiner Moderate	Later Joiner Transient	57.77	0.0000
classes	5	1362	Later Joiner Sustained	Later Joiner Transient	417.22	0.0000

Table 13: Significant Holm-Bonferroni Pairwise results for methods and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
methods	1	16434	Founder Moderate	Founder Transient	21.43	0.0001
methods	1	16434	Founder Moderate	Later Joiner Transient	22.81	0.0000
methods	1	16434	Founder Sustained	Founder Transient	7.75	0.0003
methods	1	16434	Founder Sustained	Later Joiner Transient	9.13	0.0000
methods	1	16434	Founder Transient	Later Joiner Moderate	-5.09	0.0000
methods	1	16434	Founder Transient	Later Joiner Sustained	-21.24	0.0003
methods	1	16434	Founder Transient	Later Joiner Transient	1.38	0.0000
methods	1	16434	Later Joiner Moderate	Later Joiner Transient	6.47	0.0000
methods	1	16434	Later Joiner Sustained	Later Joiner Transient	22.62	0.0000
methods	5	1362	Founder Moderate	Founder Sustained	-2489.44	0.0000
methods	5	1362	Founder Moderate	Founder Transient	547.29	0.0007
methods	5	1362	Founder Moderate	Later Joiner Moderate	308.48	0.0054
methods	5	1362	Founder Moderate	Later Joiner Sustained	-1067.11	0.0000
methods	5	1362	Founder Moderate	Later Joiner Transient	522.74	0.0000
methods	5	1362	Founder Sustained	Founder Transient	3036.73	0.0000
methods	5	1362	Founder Sustained	Later Joiner Moderate	2797.92	0.0000
methods	5	1362	Founder Sustained	Later Joiner Sustained	1422.33	0.0001
methods	5	1362	Founder Sustained	Later Joiner Transient	3012.19	0.0000
methods	5	1362	Founder Transient	Later Joiner Moderate	-238.81	0.0012
methods	5	1362	Founder Transient	Later Joiner Sustained	-1614.40	0.0000
methods	5	1362	Later Joiner Moderate	Later Joiner Sustained	-1375.59	0.0000
methods	5	1362	Later Joiner Moderate	Later Joiner Transient	214.26	0.0000
methods	5	1362	Later Joiner Sustained	Later Joiner Transient	1589.85	0.0000

Table 14: Not Significant Holm-Bonferroni Pairwise results for packages and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
packages	1	16434	Founder Moderate	Founder Sustained	0.38	1.0000
packages	1	16434	Founder Moderate	Later Joiner Moderate	-0.01	1.0000
packages	1	16434	Founder Moderate	Later Joiner Sustained	0.18	1.0000
packages	1	16434	Founder Sustained	Later Joiner Moderate	-0.39	1.0000
packages	1	16434	Founder Sustained	Later Joiner Sustained	-0.20	1.0000
packages	1	16434	Founder Transient	Later Joiner Transient	0.05	1.0000
packages	1	16434	Later Joiner Moderate	Later Joiner Sustained	0.19	1.0000
packages	5	1362	Founder Moderate	Later Joiner Moderate	-0.54	1.0000
packages	5	1362	Founder Sustained	Later Joiner Sustained	-0.46	1.0000
packages	5	1362	Founder Transient	Later Joiner Transient	-1.76	1.0000
packages	10	924	Founder Moderate	Founder Sustained	1.41	0.8267
packages	10	924	Founder Moderate	Later Joiner Moderate	-1.61	0.1575
packages	10	924	Founder Moderate	Later Joiner Sustained	-0.63	0.8267
packages	10	924	Founder Sustained	Later Joiner Sustained	-2.03	0.3229

Table 15: Not Significant Holm-Bonferroni Pairwise results for classes and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	1	16434	Founder Moderate	Founder Sustained	4.44	1.0000
classes	1	16434	Founder Moderate	Later Joiner Moderate	4.93	1.0000
classes	1	16434	Founder Moderate	Later Joiner Sustained	2.39	0.7561
classes	1	16434	Founder Sustained	Later Joiner Moderate	0.49	1.0000
classes	1	16434	Founder Sustained	Later Joiner Sustained	-2.05	0.7561
classes	1	16434	Later Joiner Moderate	Later Joiner Sustained	-2.54	0.7561
classes	5	1362	Founder Transient	Later Joiner Transient	-6.89	0.9533
classes	10	924	Founder Moderate	Founder Sustained	-7.66	1.0000
classes	10	924	Founder Moderate	Later Joiner Moderate	18.61	0.5792
classes	10	924	Founder Moderate	Later Joiner Sustained	12.95	0.5163
classes	10	924	Founder Sustained	Later Joiner Moderate	26.27	0.9912
classes	10	924	Founder Sustained	Later Joiner Sustained	20.61	0.9912
classes	10	924	Later Joiner Moderate	Later Joiner Sustained	-5.66	1.0000
classes	20	237	Founder Sustained	Later Joiner Sustained	25.59	0.2857

Table 16: Not Significant Holm-Bonferroni Pairwise results for methods and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
methods	1	16434	Founder Moderate	Founder Sustained	13.68	1.0000
methods	1	16434	Founder Moderate	Later Joiner Moderate	16.34	1.0000
methods	1	16434	Founder Moderate	Later Joiner Sustained	0.19	0.6748
methods	1	16434	Founder Sustained	Later Joiner Moderate	2.66	1.0000
methods	1	16434	Founder Sustained	Later Joiner Sustained	-13.49	0.9405
methods	1	16434	Later Joiner Moderate	Later Joiner Sustained	-16.15	0.5367
methods	5	1362	Founder Transient	Later Joiner Transient	-24.55	0.8554
methods	10	924	Founder Moderate	Founder Sustained	13.47	1.0000
methods	10	924	Founder Moderate	Later Joiner Moderate	119.34	0.6249
methods	10	924	Founder Moderate	Later Joiner Sustained	86.42	0.6249
methods	10	924	Founder Sustained	Later Joiner Moderate	105.87	0.7351
methods	10	924	Founder Sustained	Later Joiner Sustained	72.95	0.7351
methods	10	924	Later Joiner Moderate	Later Joiner Sustained	-32.92	1.0000
methods	20	237	Founder Sustained	Later Joiner Sustained	106.14	0.5131

Post-hoc Pairwise Analysis: Holm-Bonferroni Correction

To evaluate specific distribution differences between isolated developer interaction groups without violating normality or homoscedastic constraints, we implement a two-sample rank-order framework adjusted by the sequentially rejective Holm method.

The raw P-values for a set of m concurrent hypotheses are sorted in ascending order:

$$P_{(1)} \leq P_2 \leq \ldots \leq P_{(m)} \tag{14}$$

Each index i $(1 \leq i \leq m)$ is evaluated sequentially against a dynamically scaled significance threshold. The condition for rejecting the local null hypothesis $H_{(i)}$ is defined as:

$$P_{(i)} \leq \frac{\alpha}{m - i + 1} \tag{15}$$

Where:

- α is the nominal family-wise error criteria cutoff (configured to 0.05).
- m is the total number of simultaneous pairwise comparisons evaluated.
- i is the ordered ranking position of the active sub-test segment.

The sequence terminates at the first instance where $P_{(i)} > \frac{\alpha}{m-i+1}$, retaining all remaining null structures.

12 Strategy logistic regression - For all components touched for all commits

To identify early indicators of long-term developer engagement, a Logistic Regression model was employed. The model treats the developer category (Sustained vs. Non-Sustained) as a binary dependent variable. The following metrics are reported:

- **Coefficient (β):** This value represents the change in the log-odds of a developer becoming 'Sustained' for every unit increase in the component count. A positive coefficient indicates that higher activity in that component increases the likelihood of long-term retention.
- **Odds Ratio (e^{β}):** Perhaps the most interpretable metric, the Odds Ratio quantifies the relative increase in the likelihood of being a Sustained developer. For example, an Odds Ratio of 1.05 implies a 5% increase in the odds of becoming sustained for each additional component touched.
- **Std. Err (Standard Error):** Measures the accuracy of the coefficient estimate. A lower standard error relative to the coefficient indicates a more precise estimate.
- **p-Value ($P > |z|$):** Indicates the statistical significance of the predictor. In this study, a p-Value < 0.05 suggests that the specific component count is a significant predictor of whether a developer will transition into a sustained role.
- **30% Threshold:** The threshold number of components to touch to achieve a 30% empirical probability of transitioning into a sustained contributor.
- **Moderate Met Threshold:** The moderate number of developers to have met the threshold, the percentage of developers in brackets.
- **Sustained Met Threshold:** The sustained number of developers to have met the threshold, the percentage of developers in brackets.

Interpretation of Intercept vs. Component Count:

- **Intercept:** Represents the baseline log-odds of the dependent variable (Sustained Status) when the independent variable (e.g., Package Count) is zero. In this study, the consistently negative intercepts indicate that the probability of a developer becoming 'Sustained' is statistically low without active engagement in the codebase.
- **Component Count (Slope):** Represents the effect size of architectural engagement. The positive coefficients and Odds Ratios greater than 1.0 demonstrate that every additional unique component touched by a developer significantly increases their likelihood of transitioning into a sustained contributor.

Table 17: Strategy Logistic Regression for all components against number of commits for developers

Component	Number of First Commits	Type	Number of Developers	Not Sustained μ	Sustained μ	Coefficient	Odds Ratio	Std. Err	p-Value	30.0% Threshold	Not Sustained Met Threshold	Sustained Met Threshold
packages	1	Intercept	16434	0.48	5.76	-4.6270	0.0098	0.0785	0.0000	-	-	-
packages	1	Packages Count	16434	0.48	5.76	0.2433	1.2755	0.0136	0.0000	**15.5351**	38 (0.2%)	15 (6.3%)
packages	5	Intercept	1362	11.26	35.24	-2.7036	0.0670	0.1256	0.0000	-	-	-
packages	5	Packages Count	1362	11.26	35.24	0.0618	1.0638	0.0049	0.0000	**30.0232**	71 (6.3%)	100 (42.2%)
packages	10	Intercept	924	13.85	32.42	-1.9820	0.1378	0.1286	0.0000	-	-	-
packages	10	Packages Count	924	13.85	32.42	0.0463	1.0474	0.0050	0.0000	**24.5148**	104 (15.1%)	106 (44.7%)
classes	1	Classes Count	16434	1.43	36.49	0.0514	1.0528	0.0028	0.0000	**73.2126**	45 (0.3%)	28 (11.8%)
classes	5	Intercept	1362	57.53	447.83	-3.3919	0.0336	0.1557	0.0000	-	-	-
classes	5	Classes Count	1362	57.53	447.83	0.0130	1.0131	0.0009	0.0000	**195.0338**	56 (5.0%)	156 (65.8%)
classes	10	Intercept	924	75.67	407.66	-2.7164	0.0661	0.1544	0.0000	-	-	-
classes	10	Classes Count	924	75.67	407.66	0.0111	1.0112	0.0009	0.0000	**167.9633**	64 (9.3%)	158 (66.7%)
methods	1	Methods Count	16434	4.78	133.11	0.0103	1.0104	0.0007	0.0000	**353.2015**	34 (0.2%)	16 (6.8%)
methods	5	Intercept	1362	207.57	1737.49	-3.0246	0.0486	0.1351	0.0000	-	-	-
methods	5	Methods Count	1362	207.57	1737.49	0.0028	1.0028	0.0002	0.0000	**778.7300**	47 (4.2%)	144 (60.8%)
methods	10	Intercept	924	275.76	1574.80	-2.3681	0.0937	0.1350	0.0000	-	-	-
methods	10	Methods Count	924	275.76	1574.80	0.0023	1.0023	0.0002	0.0000	**651.2866**	61 (8.9%)	143 (60.3%)

Predictive Identification Model

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability P of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n \tag{16}$$

Where:

- P is the probability that the developer's tenure is "Sustained".
- β_0 is the intercept.
- X_n represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- β_n are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \tag{17}$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

Model Evaluation The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy[cite: 1639].

13 Repository

13.1 Repository: 14

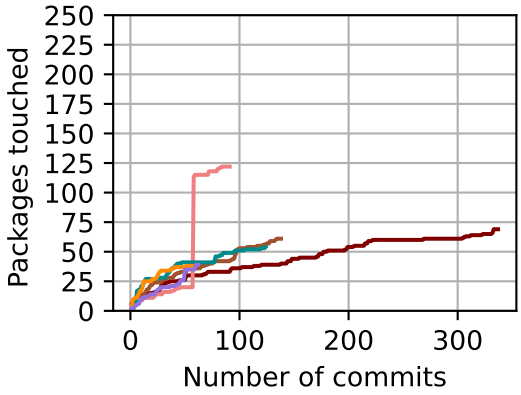
13.1.1 Time series sustained components touched by developer - 14 For developer commits and components touched.

Table 18: Table of first 6 sustained founder developers in the Scatter Developer graphs.

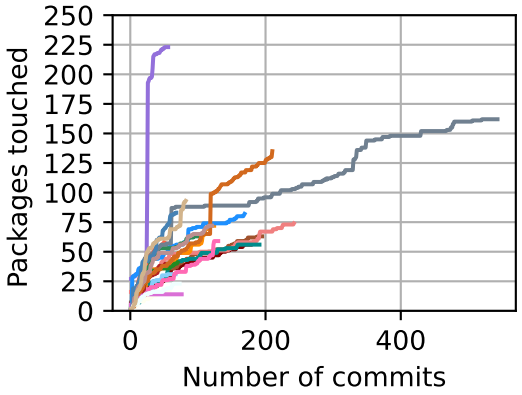
Developer ID	Colour
2930	Maroon
2932	Sienna
2934	LightCoral
2935	DarkCyan
2946	DarkOrange
2959	MediumPurple

Table 19: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

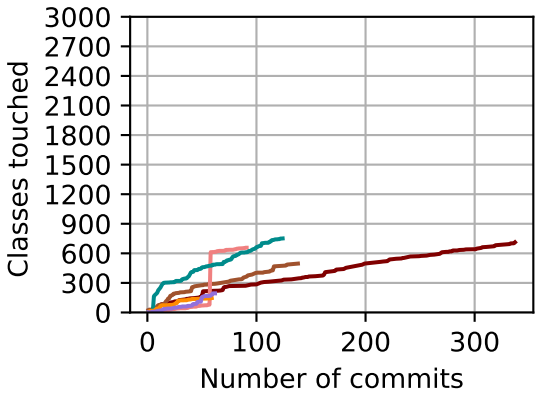
Developer ID	Colour
1774	Maroon
2986	Sienna
3000	LightCoral
3020	DarkCyan
3037	DarkOrange
3050	MediumPurple
3071	LightSkyBlue
3072	HotPink
3141	ForestGreen
3143	FireBrick
3163	DodgerBlue
3172	Tomato
3178	SeaGreen
3216	SlateGray
3217	Ivory
3272	Orchid
3339	Chocolate
3354	SteelBlue
3456	RosyBrown
3473	Azure
3485	Tan



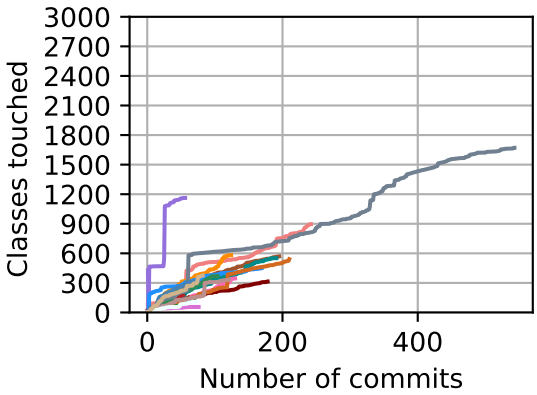
(a) Packages for sustained founder



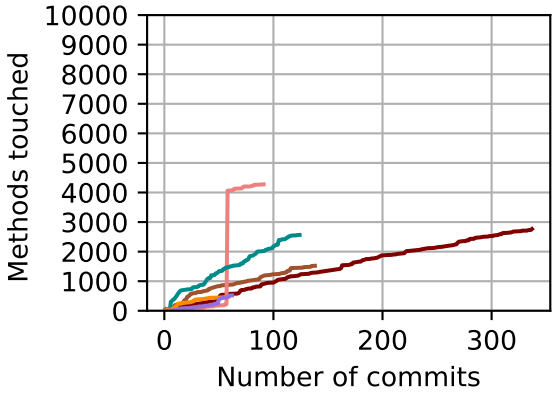
(b) Packages for sustained later joiner



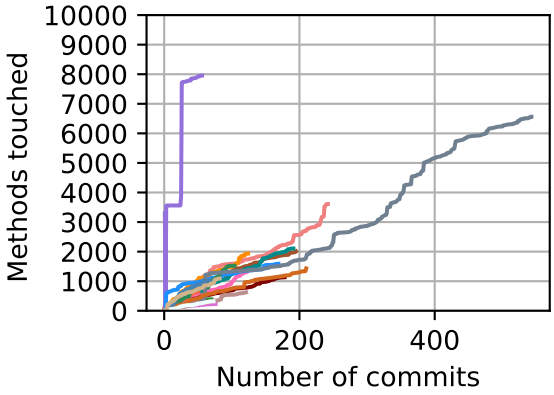
(c) Classes for sustained founder



(d) Classes for sustained later joiner



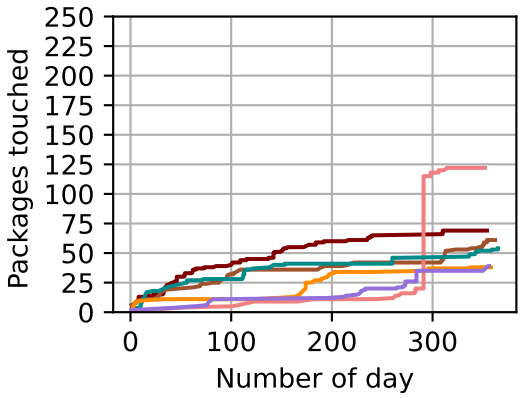
(e) Methods for sustained founder



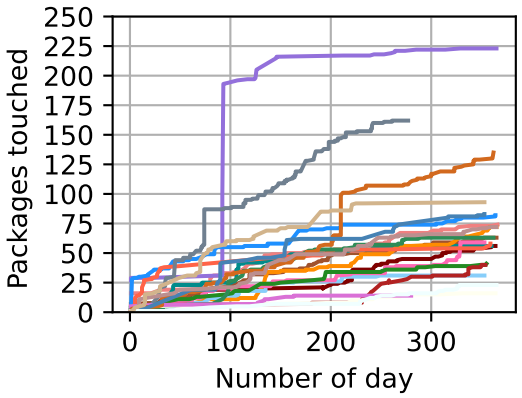
(f) Methods for sustained later joiner

Figure 28: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 18. Sustained later joiner developers with colours are shown in Table 19.

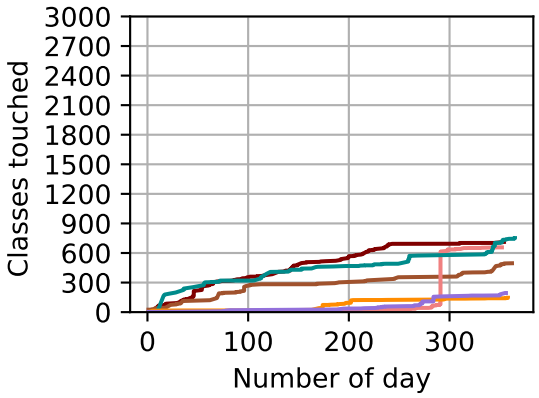
13.1.2 Time series sustained components touched by developer - 14 For developer day and components touched.



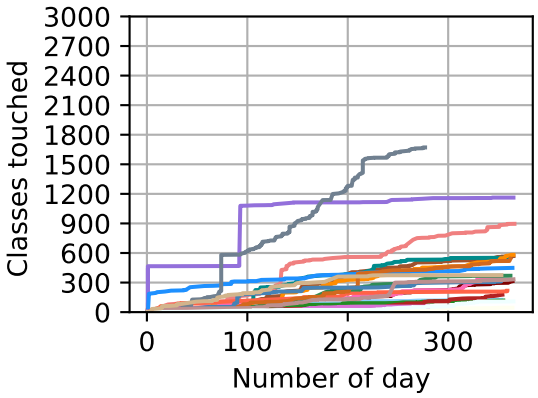
(a) Packages for sustained founder



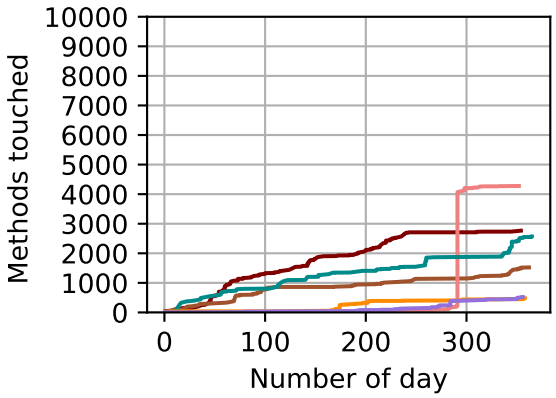
(b) Packages for sustained later joiner



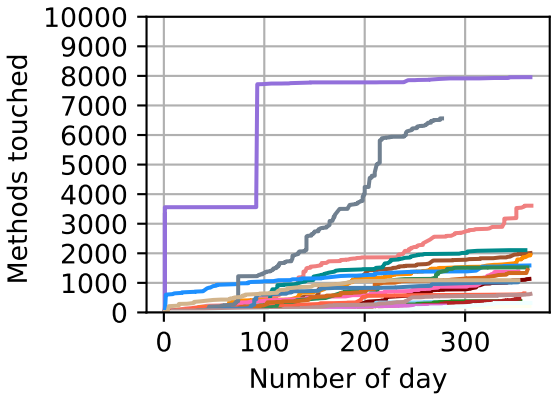
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 29: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 18. Sustained later joiner developers with colours are shown in Table 19.

13.2 Repository: 21

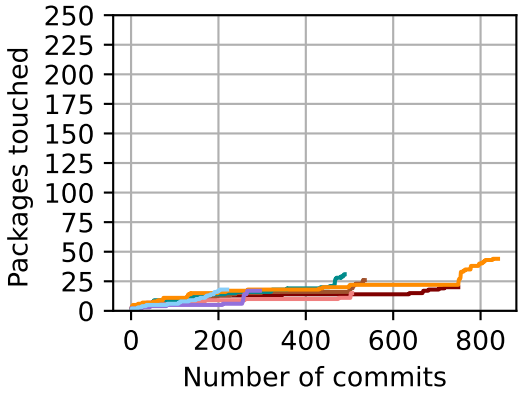
13.2.1 Time series sustained components touched by developer - 21 For developer commits and components touched.

Table 20: Table of first 7 sustained founder developers in the Scatter Developer graphs.

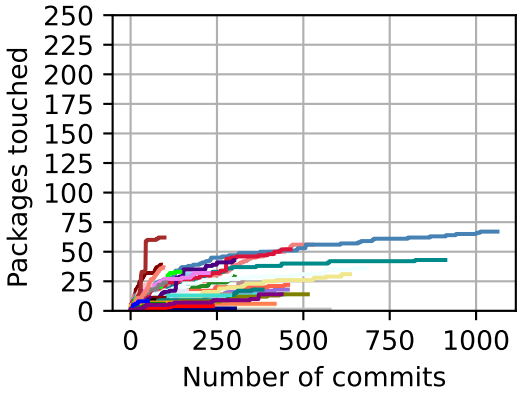
Developer ID	Colour
6245	Maroon
6246	Sienna
6247	LightCoral
6249	DarkCyan
6250	DarkOrange
6251	MediumPurple
6257	LightSkyBlue

Table 21: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

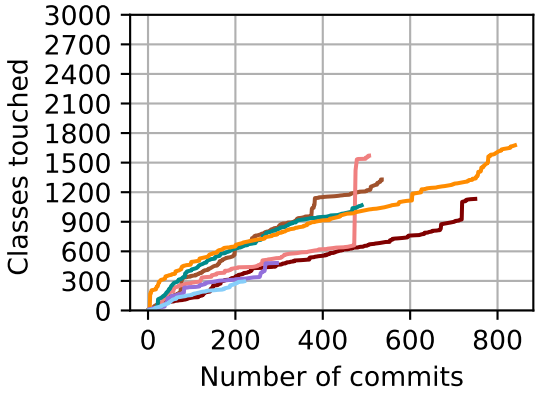
Developer ID	Colour
199	Maroon
6252	Sienna
6253	LightCoral
6259	DarkCyan
6260	DarkOrange
6262	MediumPurple
6263	LightSkyBlue
6264	HotPink
6267	ForestGreen
6271	FireBrick
6272	DodgerBlue
6276	Tomato
6278	SeaGreen
6280	SlateGray
6282	Ivory
6284	Orchid
6288	Chocolate
6295	SteelBlue
6297	RosyBrown
6298	Azure
6302	Tan
6306	Plum
6309	Crimson
6312	Khaki
6313	Lavender
6314	Indigo
6315	Turquoise
6326	Beige
6329	Silver
6333	Gold
6338	Coral
6342	Lime
6343	Maroon
6349	Navy
6352	Teal
6355	Olive
6363	Violet
6368	Gray
6369	Pink
6370	Brown
6373	Black
6374	Yellow
6375	Magenta
6379	Cyan
6391	Salmon
6410	Orange
6411	Red
6417	Green
6418	Blue
6420	Purple



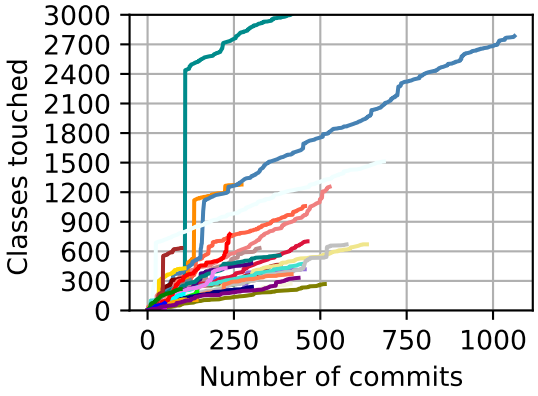
(a) Packages for sustained founder



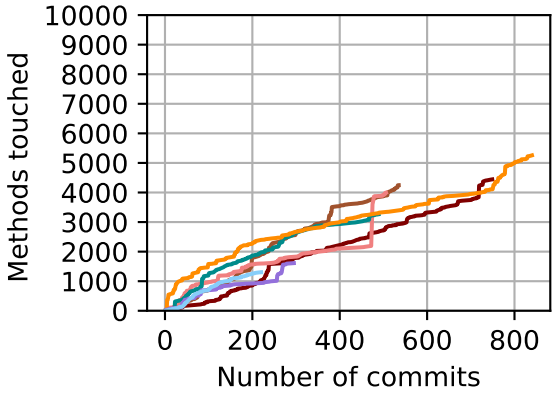
(b) Packages for sustained later joiner



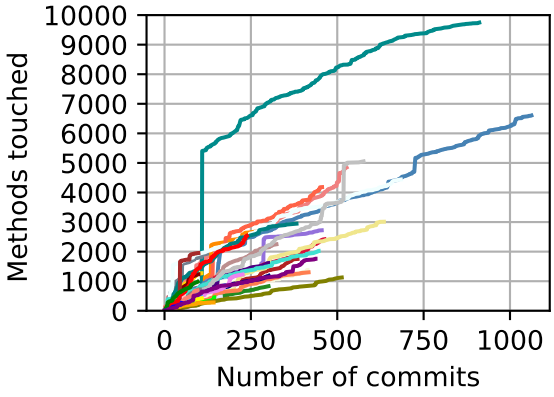
(c) Classes for sustained founder



(d) Classes for sustained later joiner



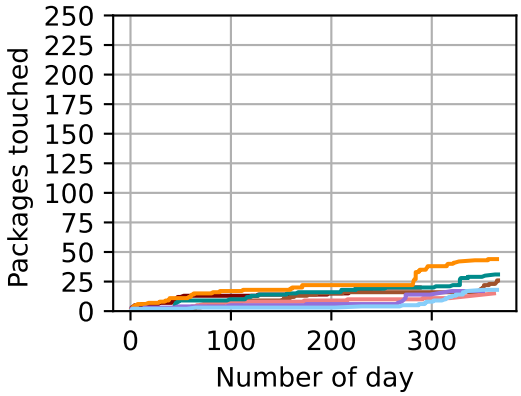
(e) Methods for sustained founder



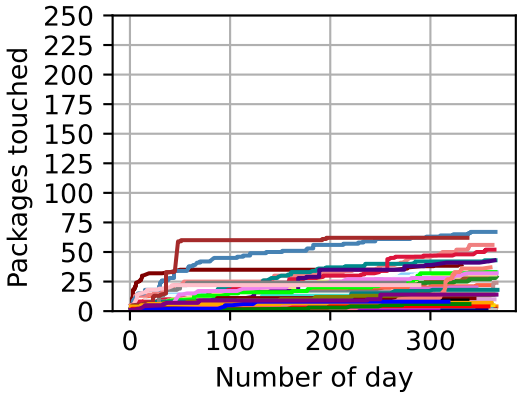
(f) Methods for sustained later joiner

Figure 30: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 20. Sustained later joiner developers with colours are shown in Table 21.

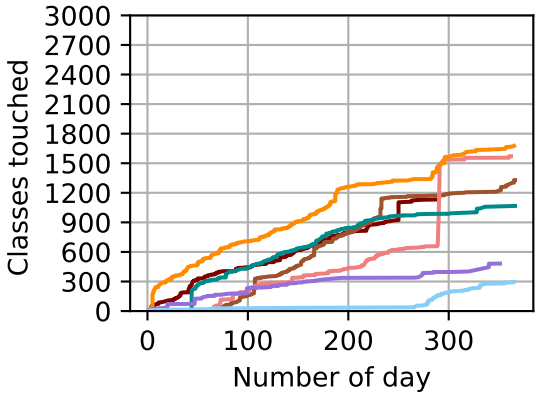
13.2.2 Time series sustained components touched by developer - 21 For developer day and components touched.



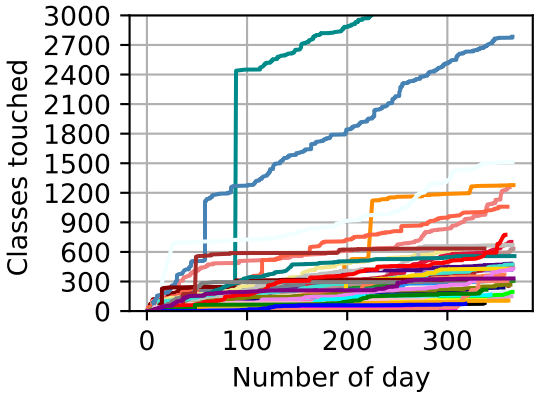
(a) Packages for sustained founder



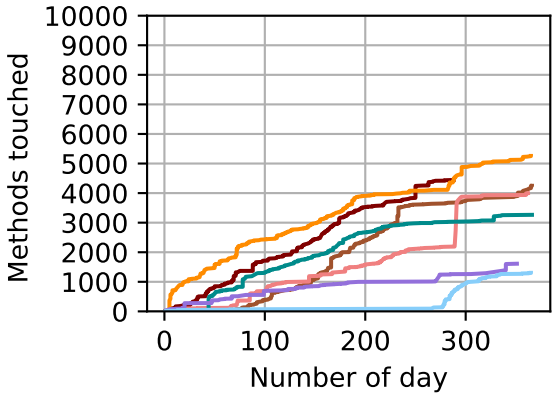
(b) Packages for sustained later joiner



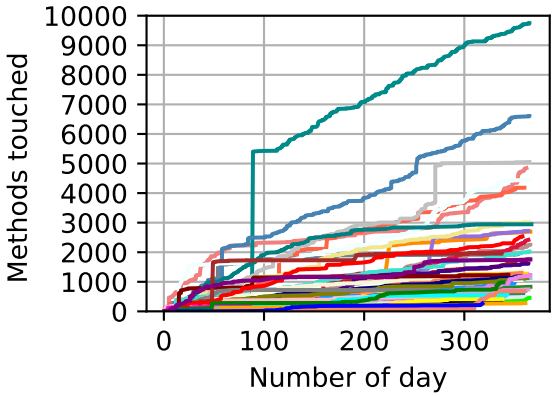
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 31: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 20. Sustained later joiner developers with colours are shown in Table 21.